

BFS Traversal

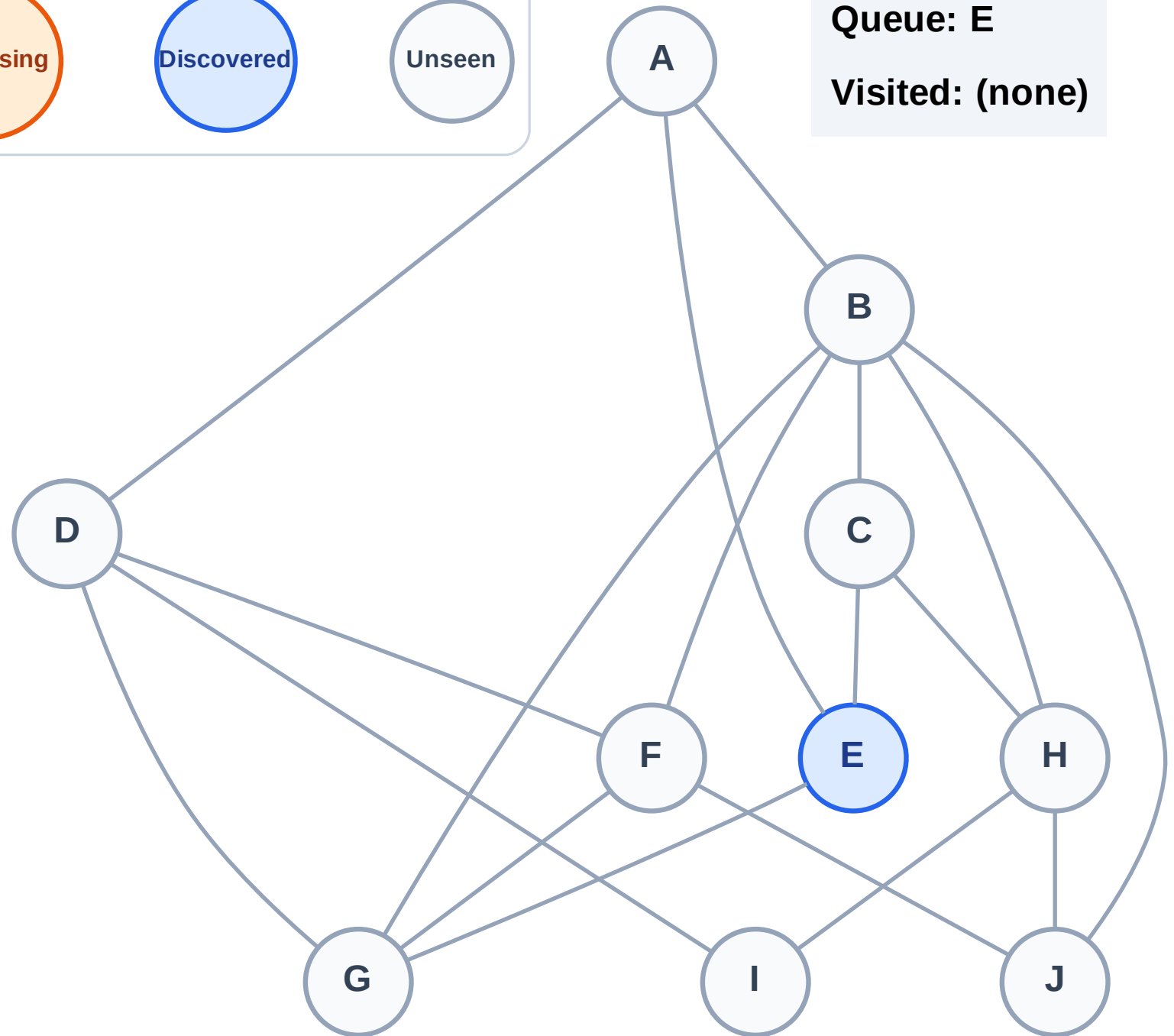
Step 1: Start at E

Legend



Queue: E

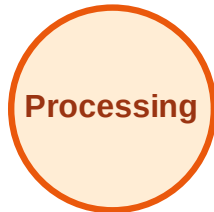
Visited: (none)



BFS Traversal

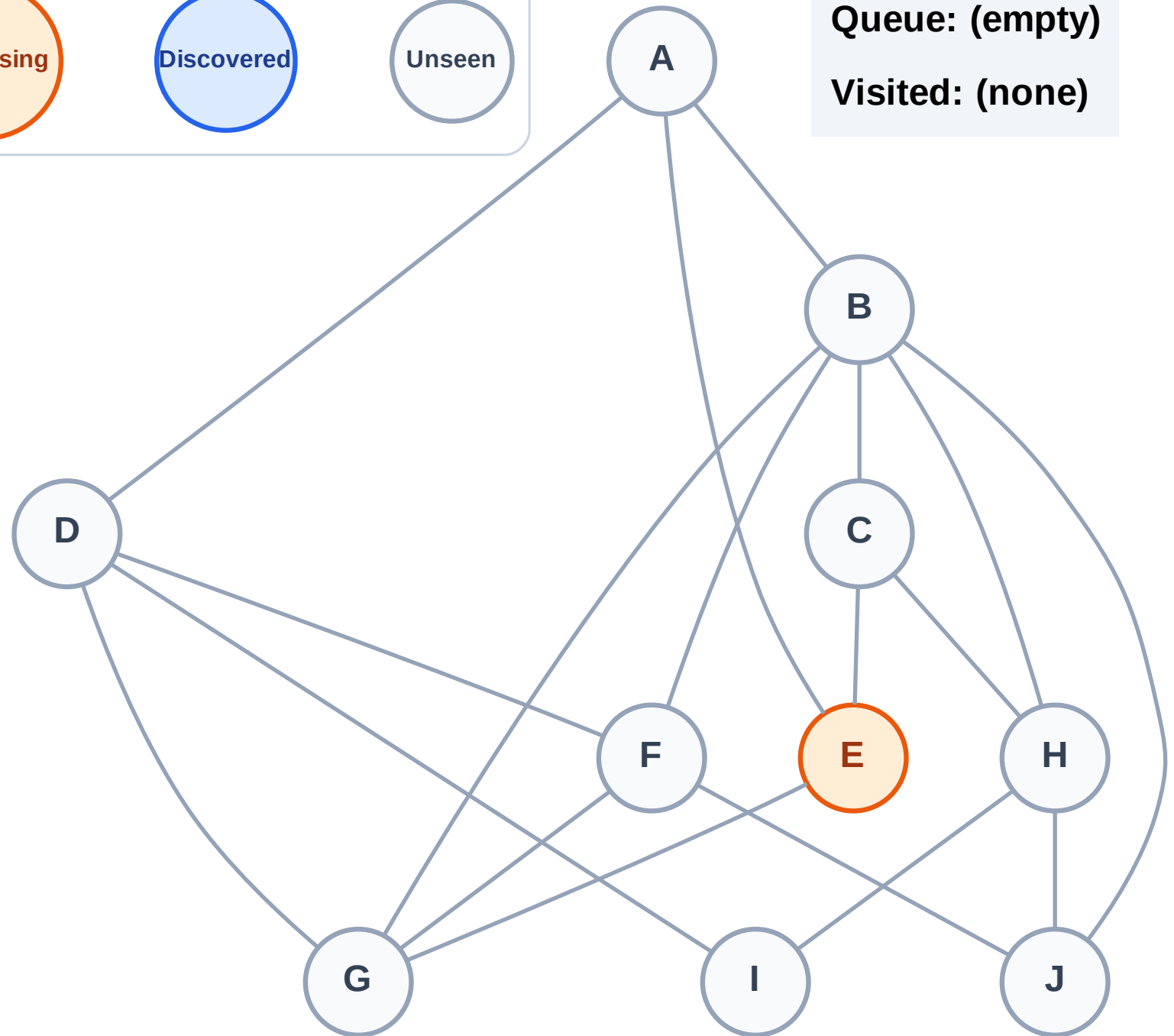
Step 2: Process node E

Legend



Queue: (empty)

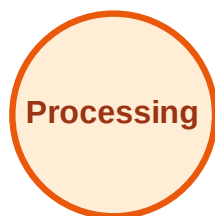
Visited: (none)



BFS Traversal

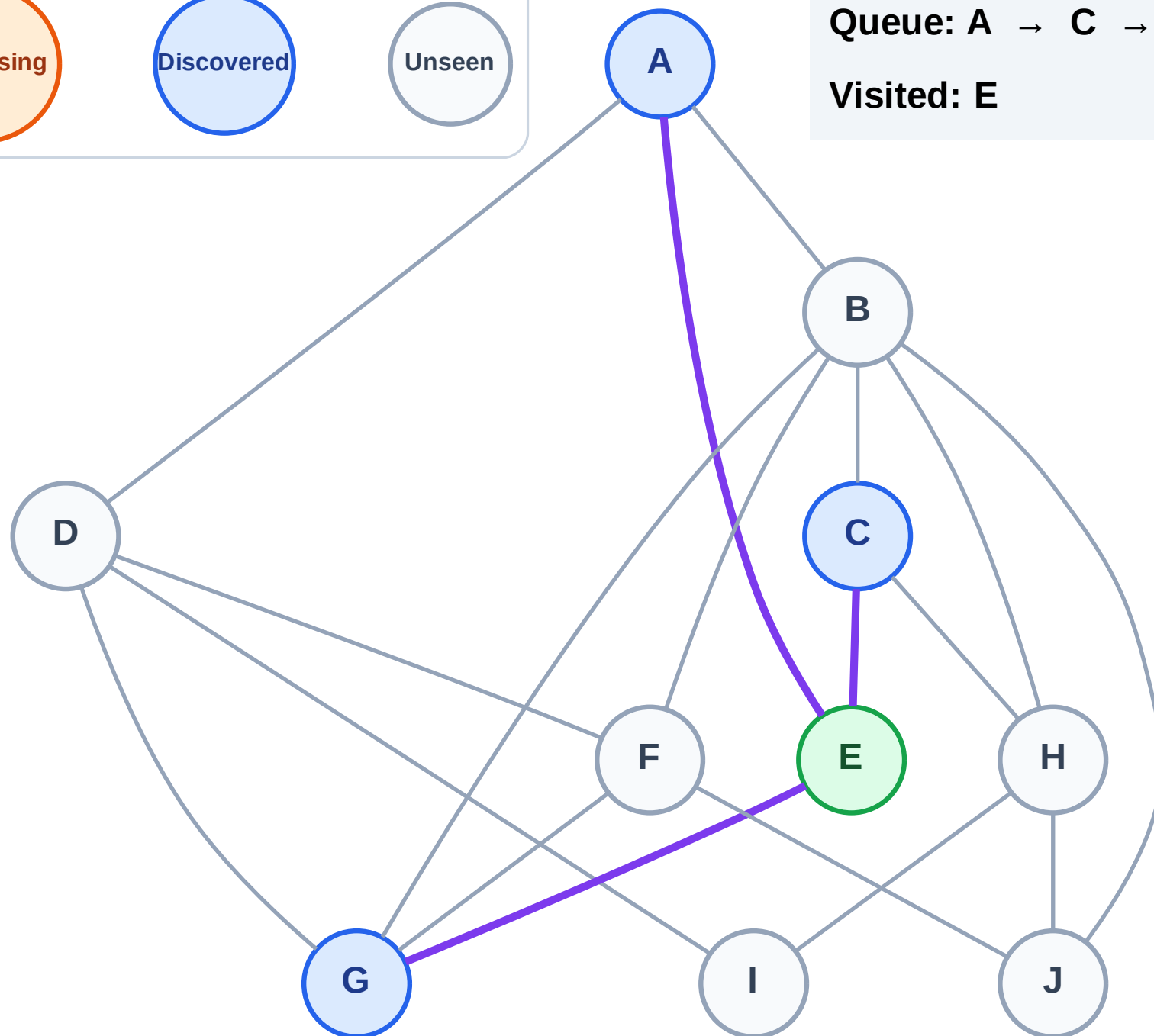
Step 3: Discover A, C, G from E

Legend



Queue: A → C → G

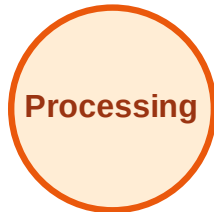
Visited: E



BFS Traversal

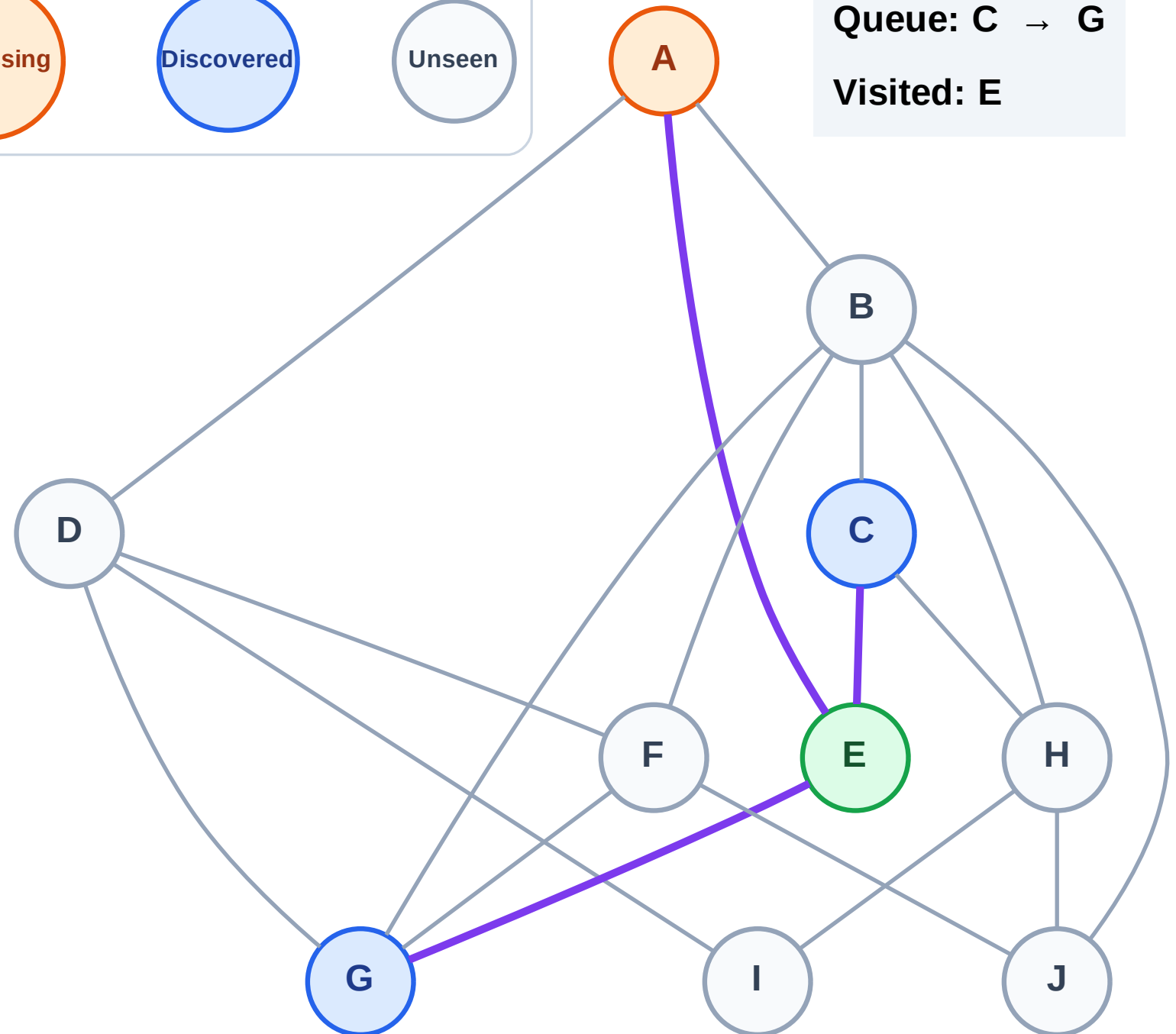
Step 4: Process node A

Legend



Queue: C → G

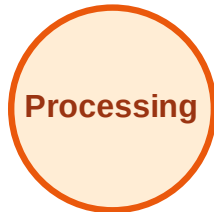
Visited: E



BFS Traversal

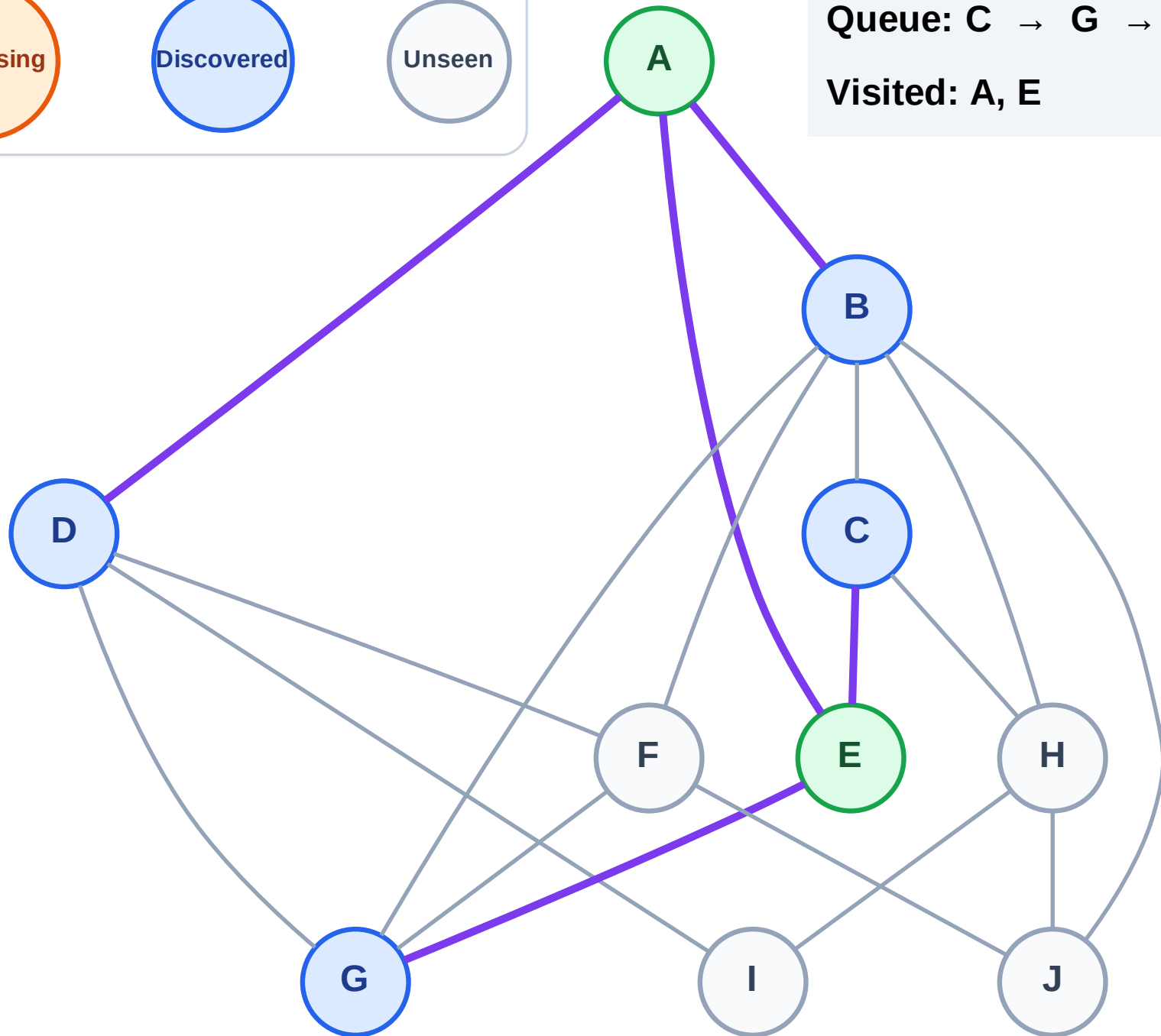
Step 5: Discover B, D from A

Legend



Queue: C → G → B → D

Visited: A, E



BFS Traversal

Step 6: Process node C

Legend

Complete

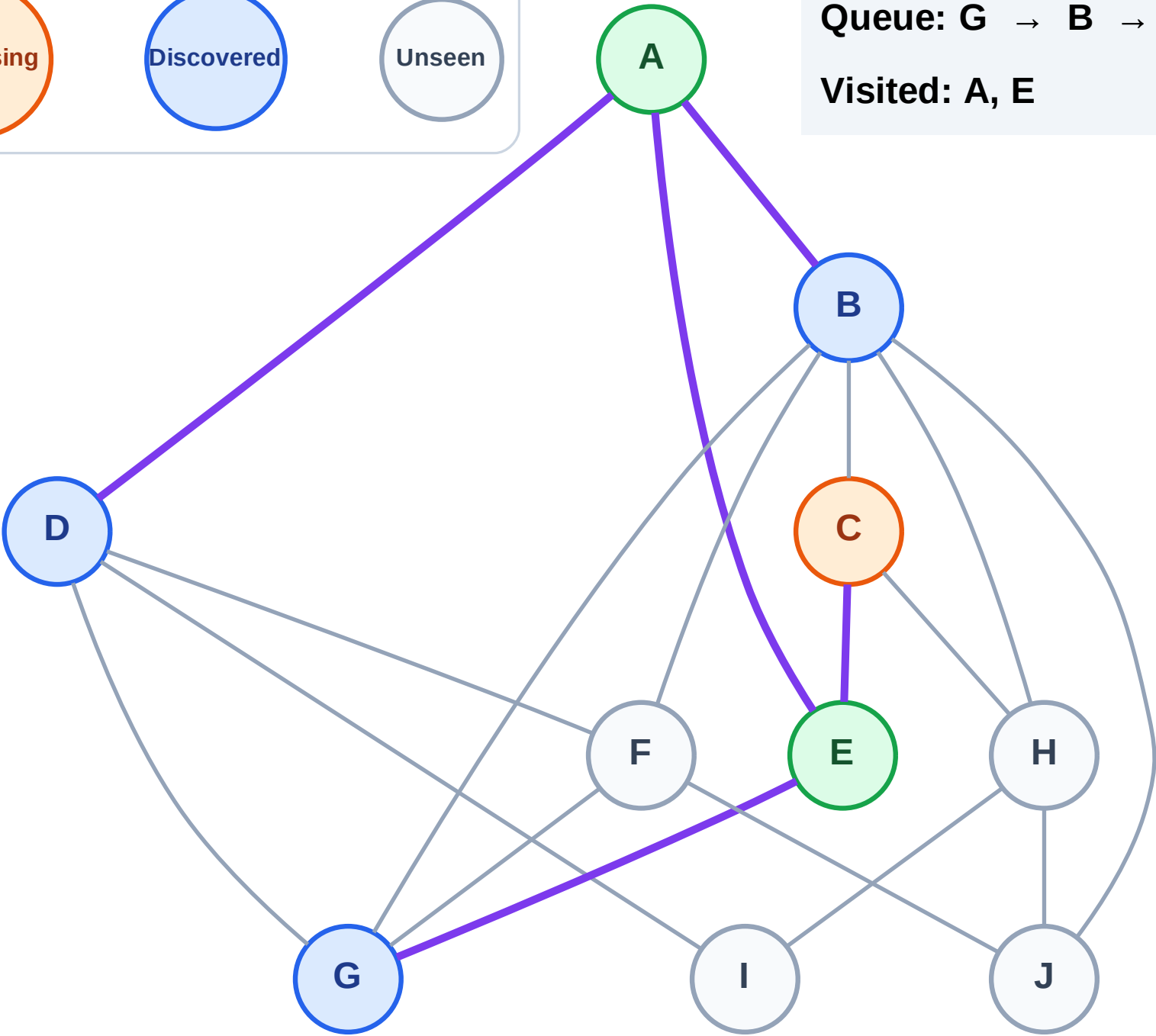
Processing

Discovered

Unseen

Queue: G → B → D

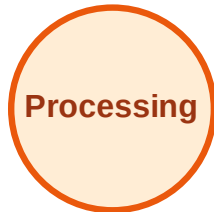
Visited: A, E



BFS Traversal

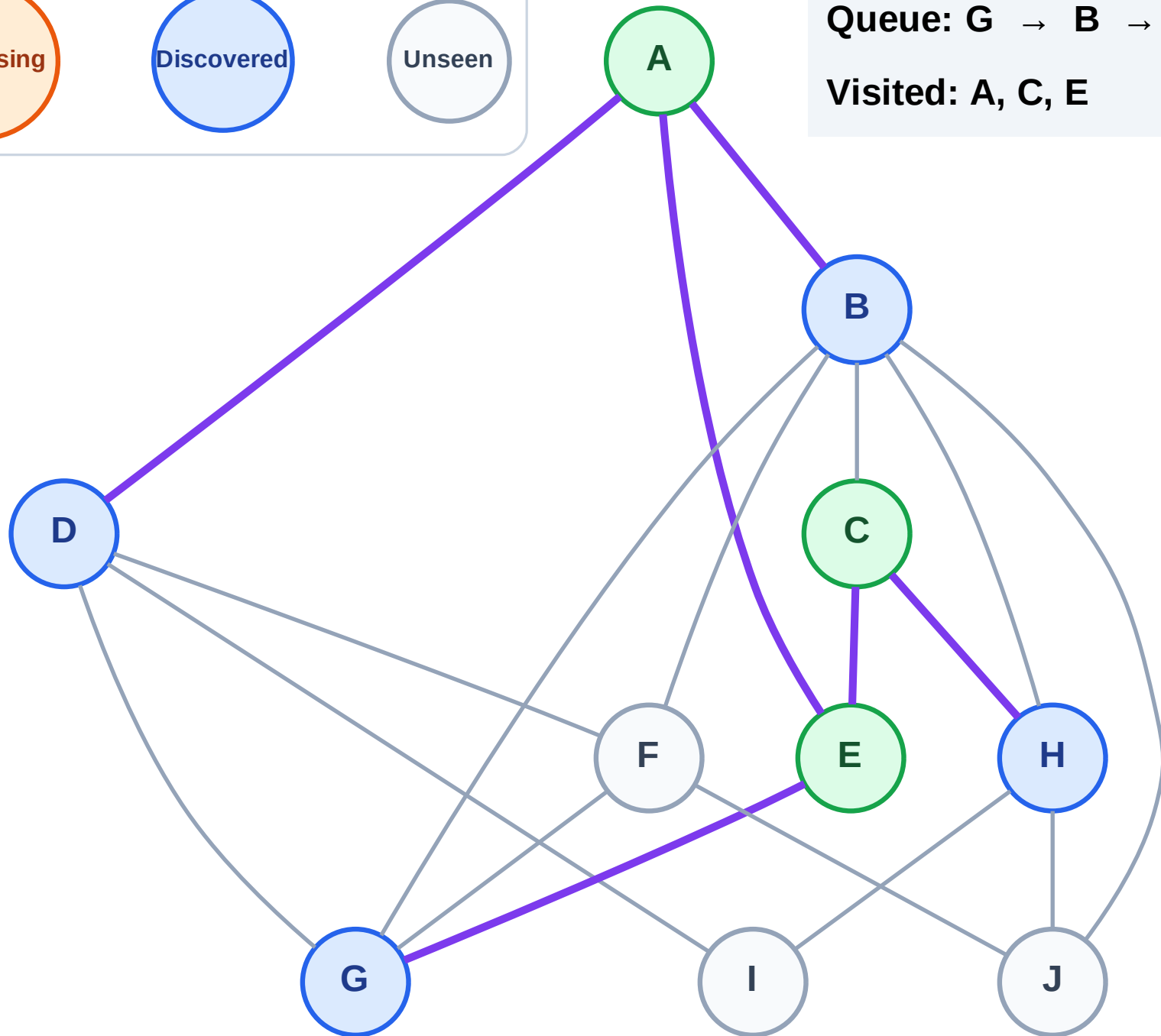
Step 7: Discover H from C

Legend



Queue: G → B → D → H

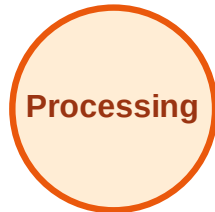
Visited: A, C, E



BFS Traversal

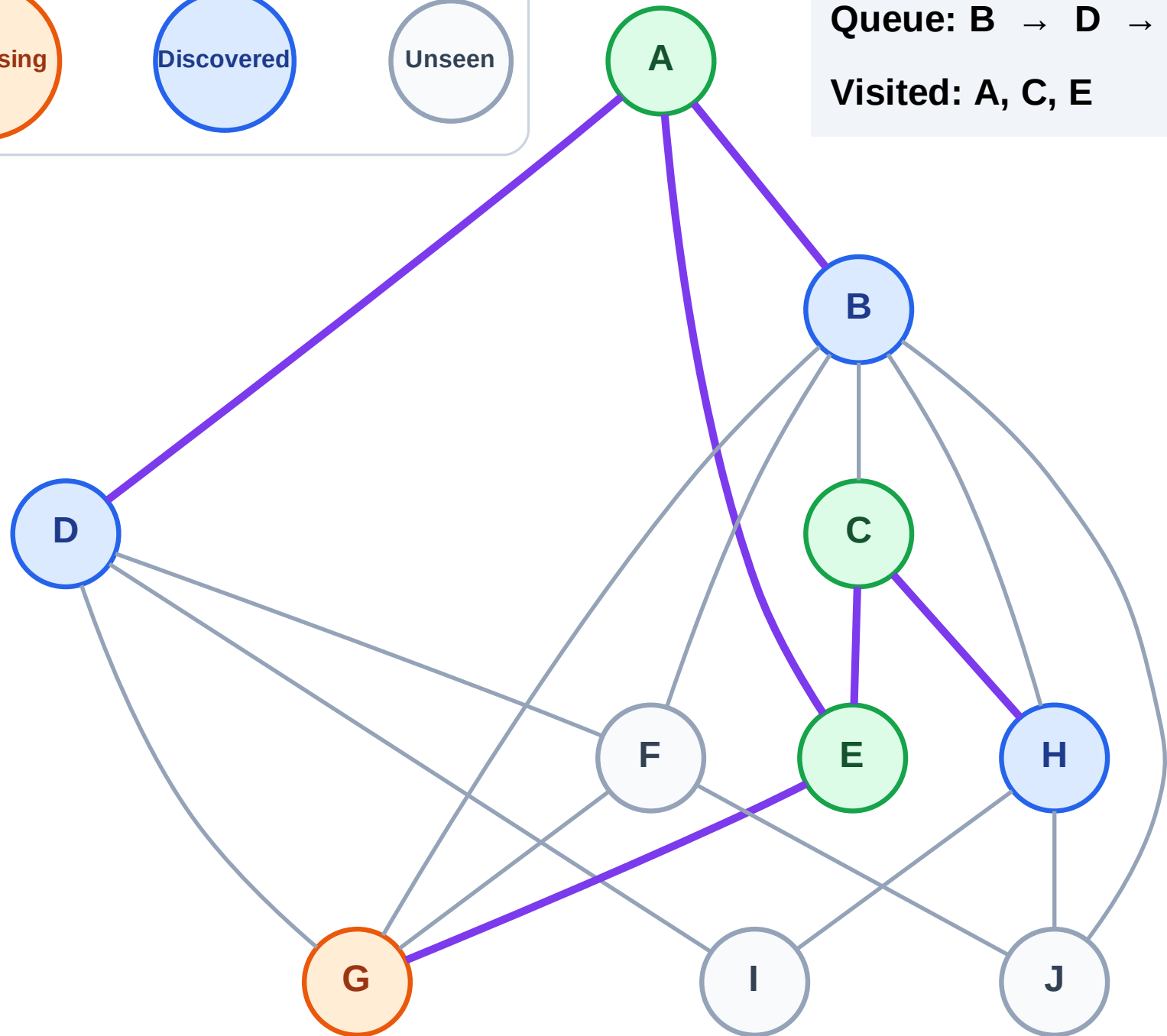
Step 8: Process node G

Legend



Queue: B → D → H

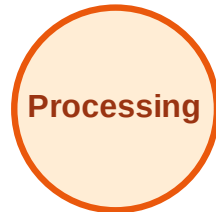
Visited: A, C, E



BFS Traversal

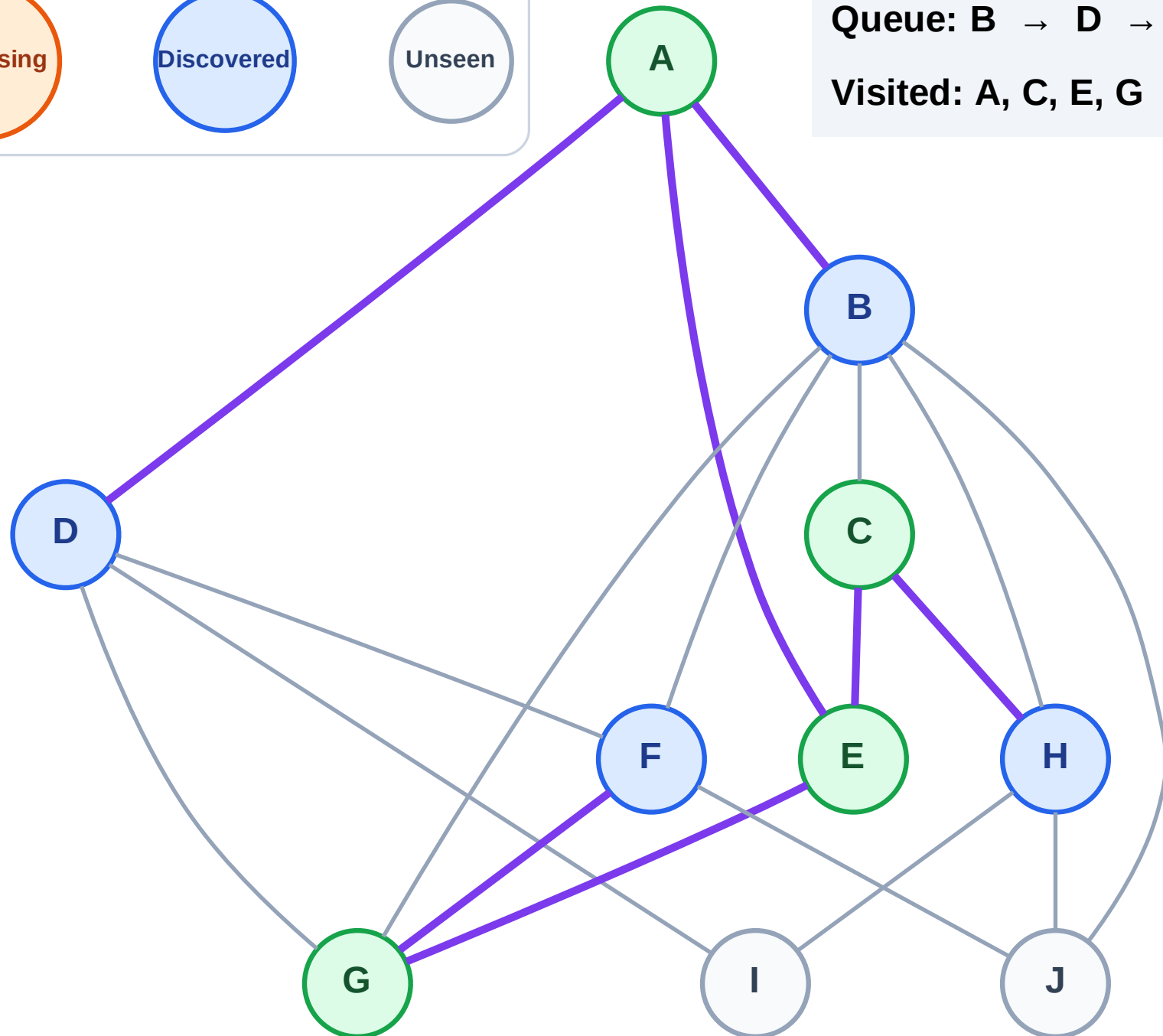
Step 9: Discover F from G

Legend



Queue: B → D → H → F

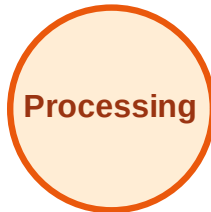
Visited: A, C, E, G



BFS Traversal

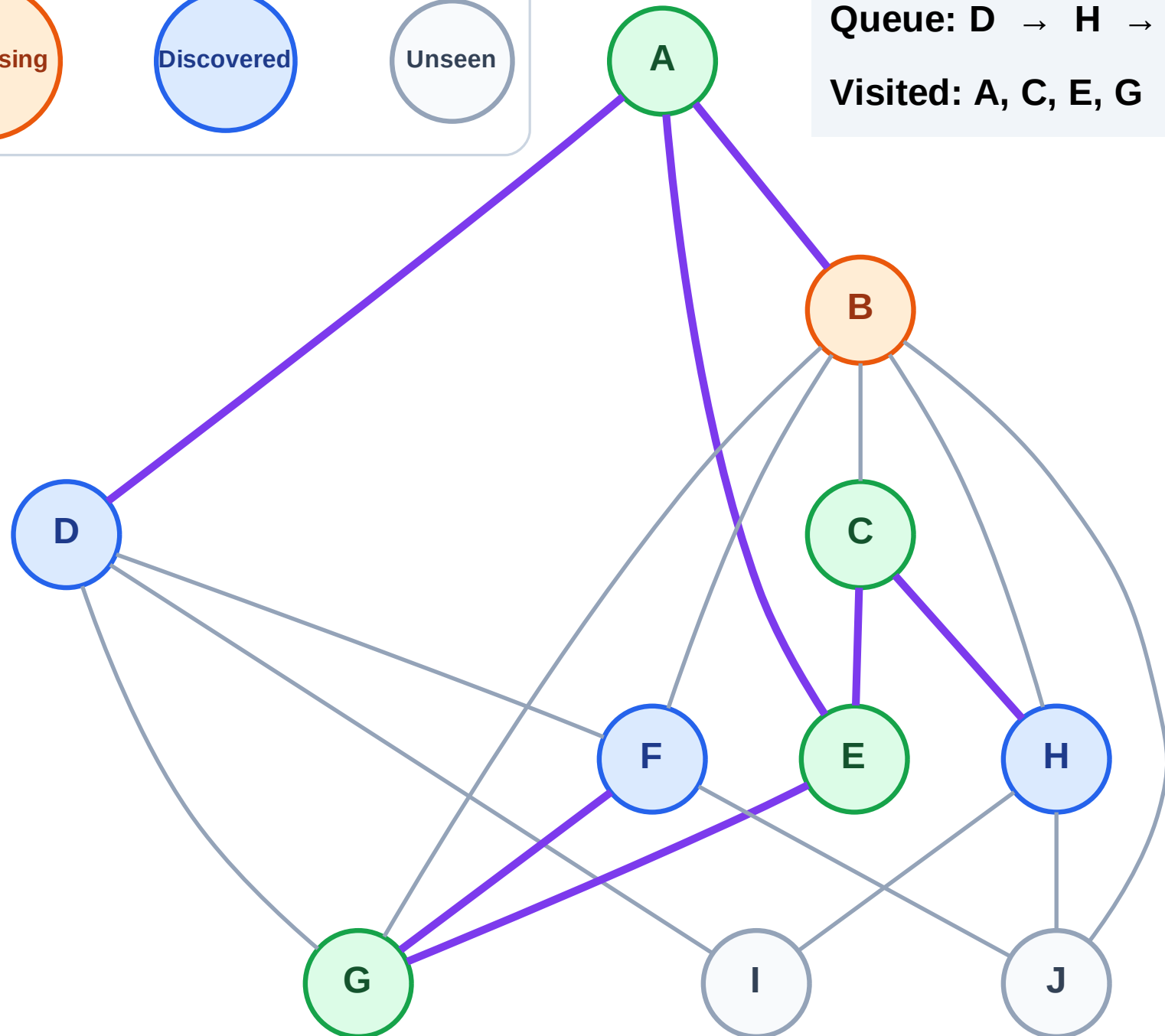
Step 10: Process node B

Legend



Queue: D → H → F

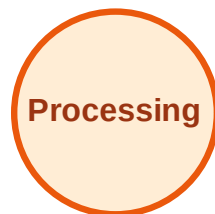
Visited: A, C, E, G



BFS Traversal

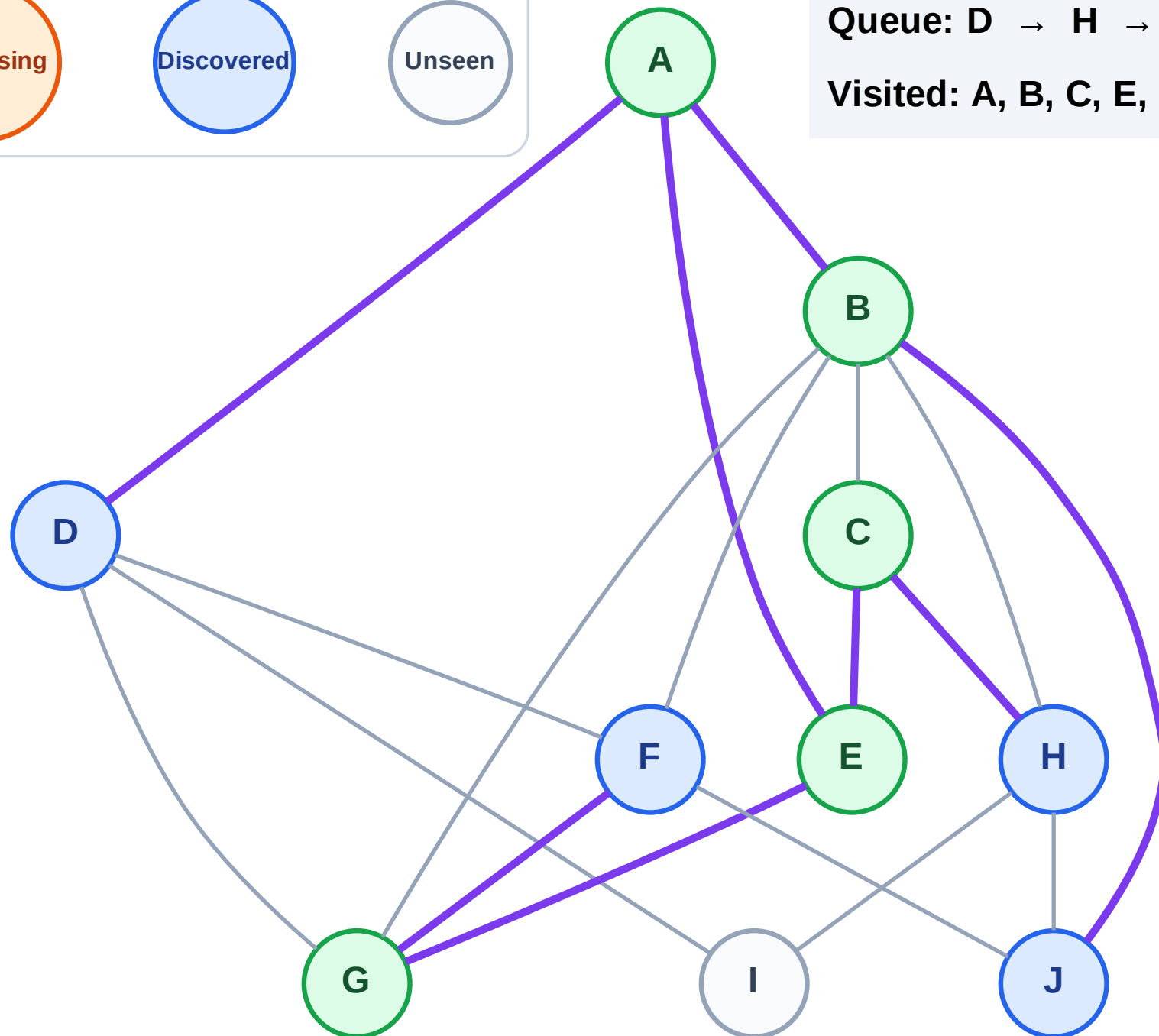
Step 11: Discover J from B

Legend



Queue: D → H → F → J

Visited: A, B, C, E, G



BFS Traversal

Step 12: Process node D

Legend

Complete

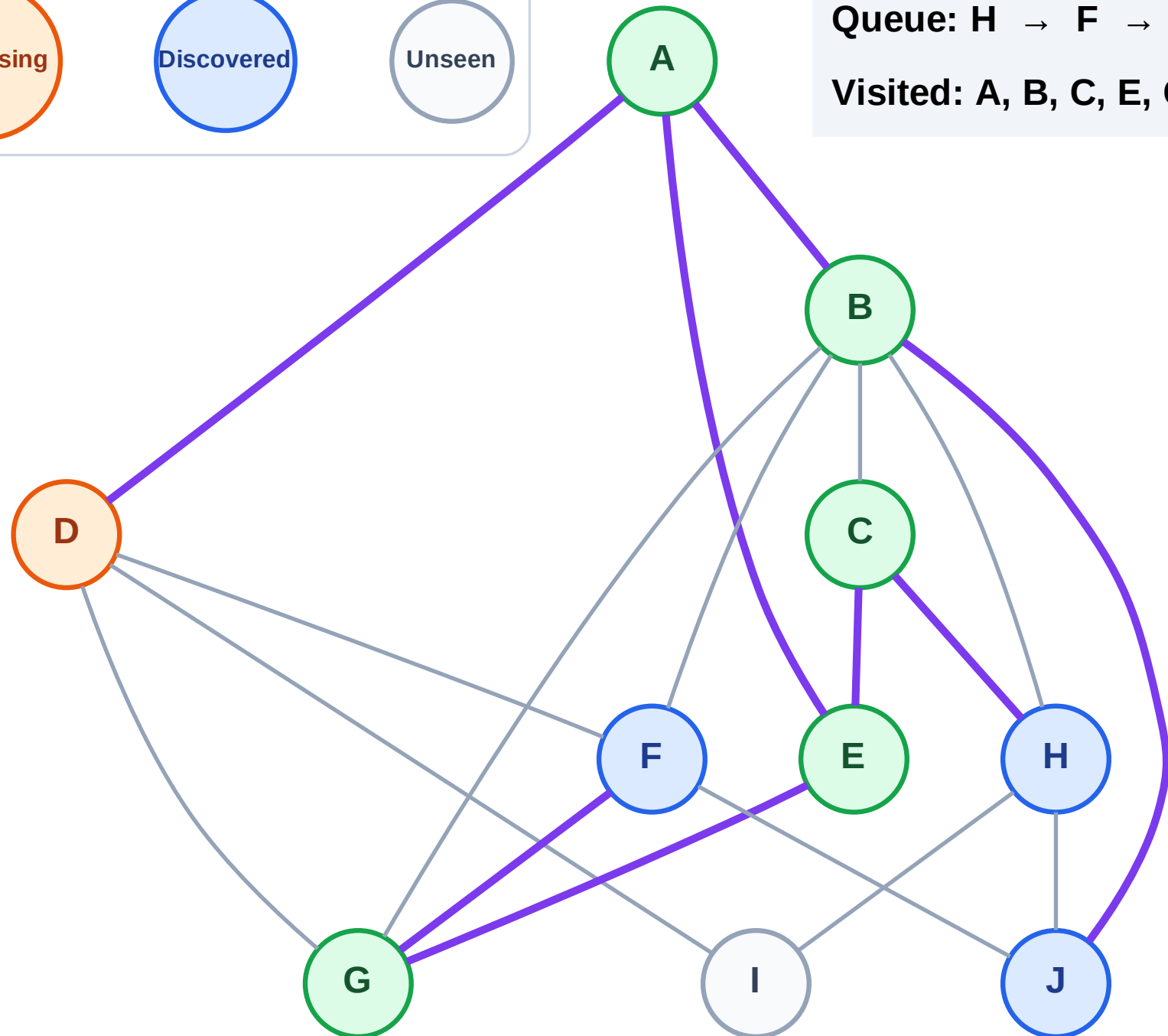
Processing

Discovered

Unseen

Queue: H → F → J

Visited: A, B, C, E, G



BFS Traversal

Step 13: Discover I from D

Legend

Complete

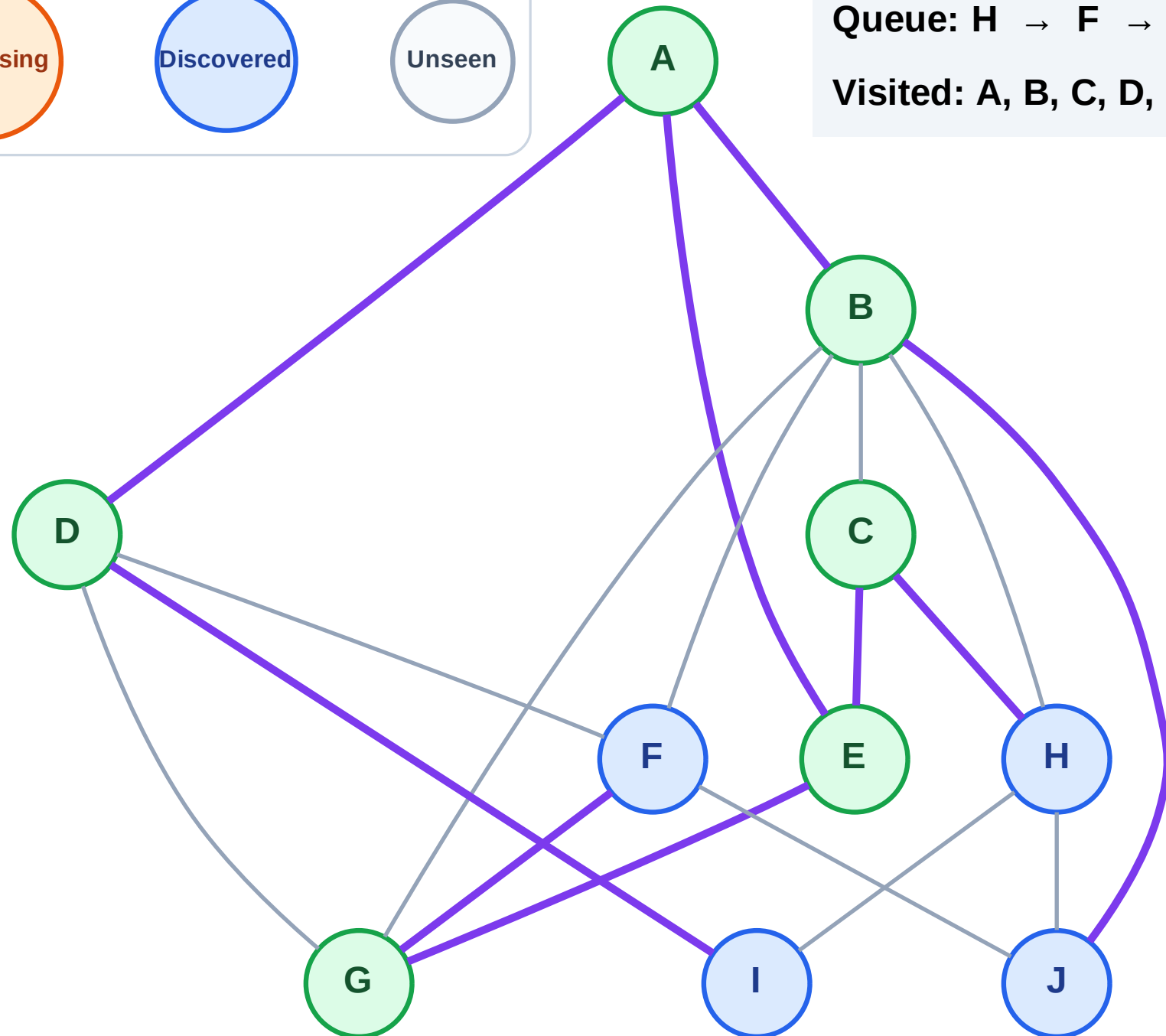
Processing

Discovered

Unseen

Queue: H → F → J → I

Visited: A, B, C, D, E, G



BFS Traversal

Step 14: Process node H

Legend

Complete

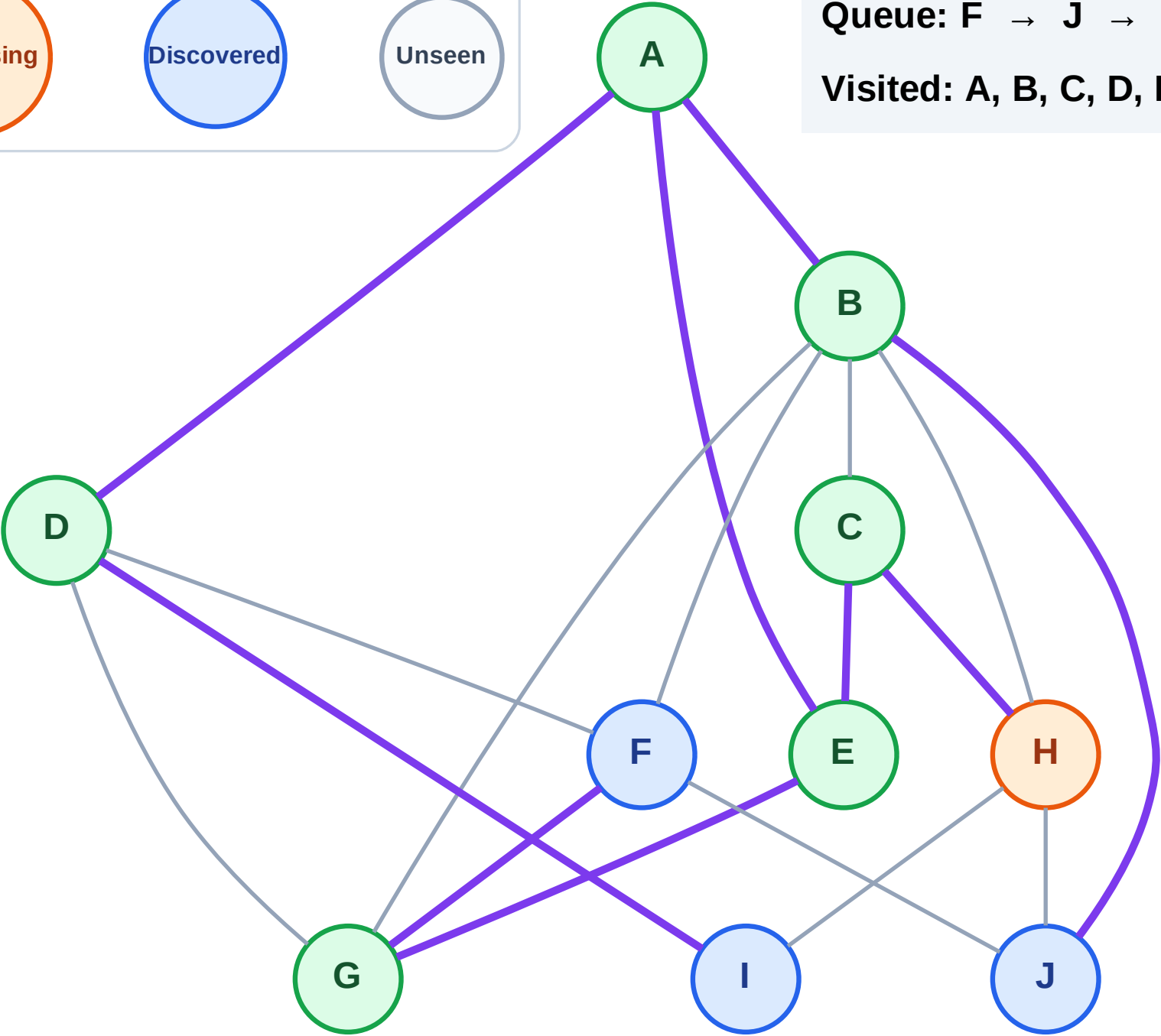
Processing

Discovered

Unseen

Queue: F → J → I

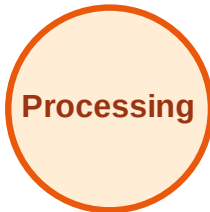
Visited: A, B, C, D, E, G



BFS Traversal

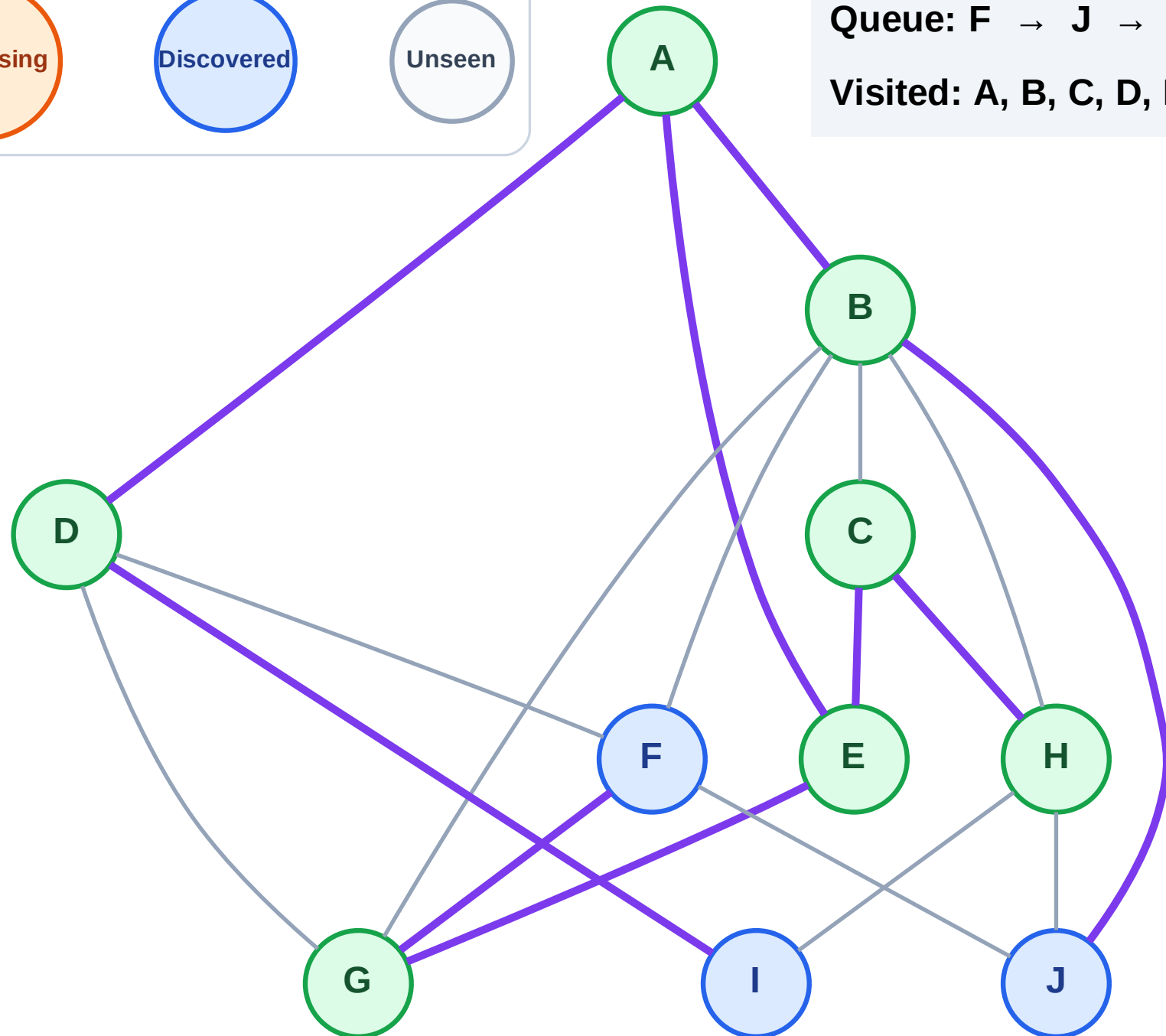
Step 15: No new neighbors from H

Legend



Queue: F → J → I

Visited: A, B, C, D, E, G, H



BFS Traversal

Step 16: Process node F

Legend

Complete

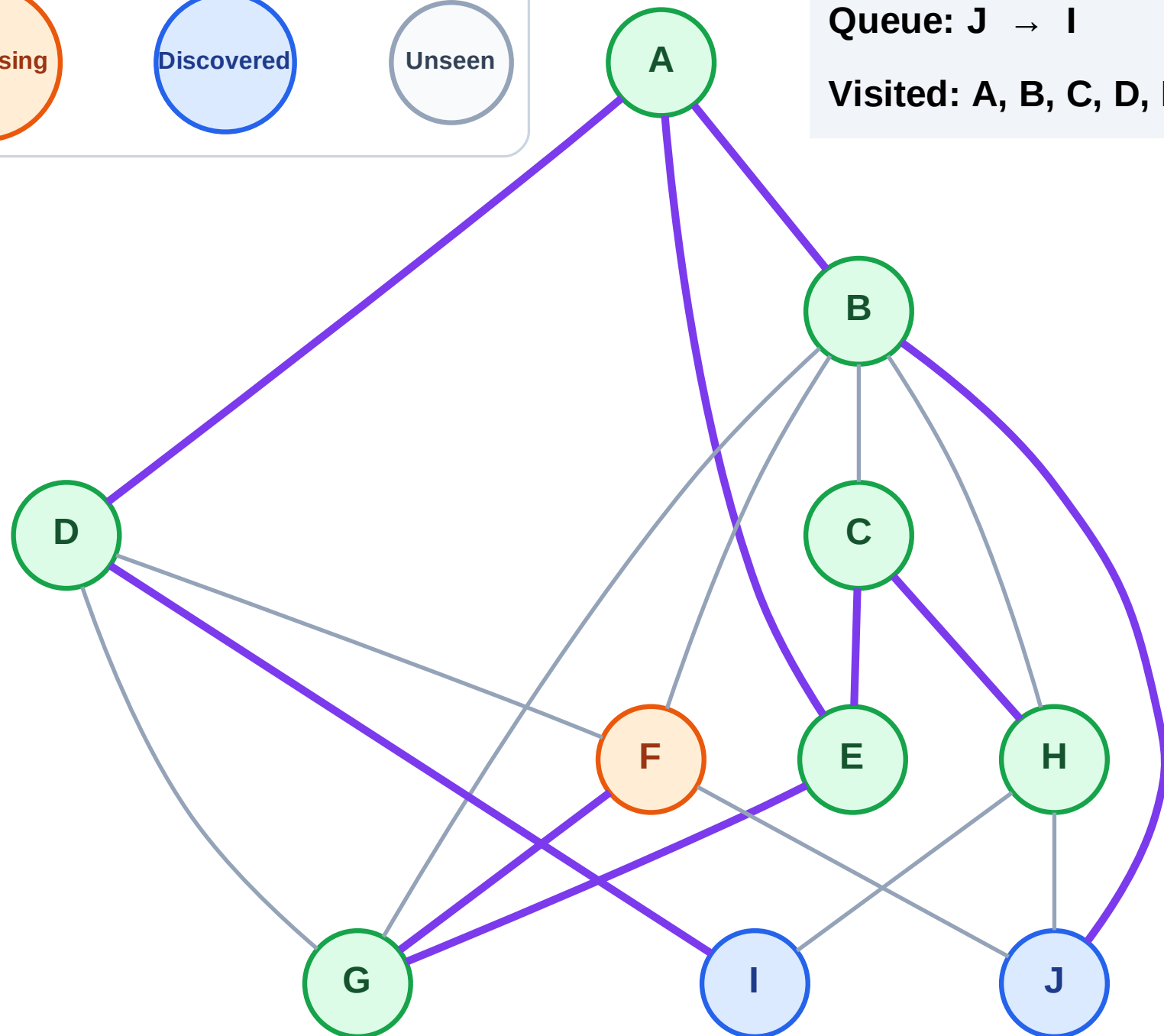
Processing

Discovered

Unseen

Queue: J → I

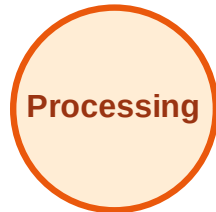
Visited: A, B, C, D, E, G, H



BFS Traversal

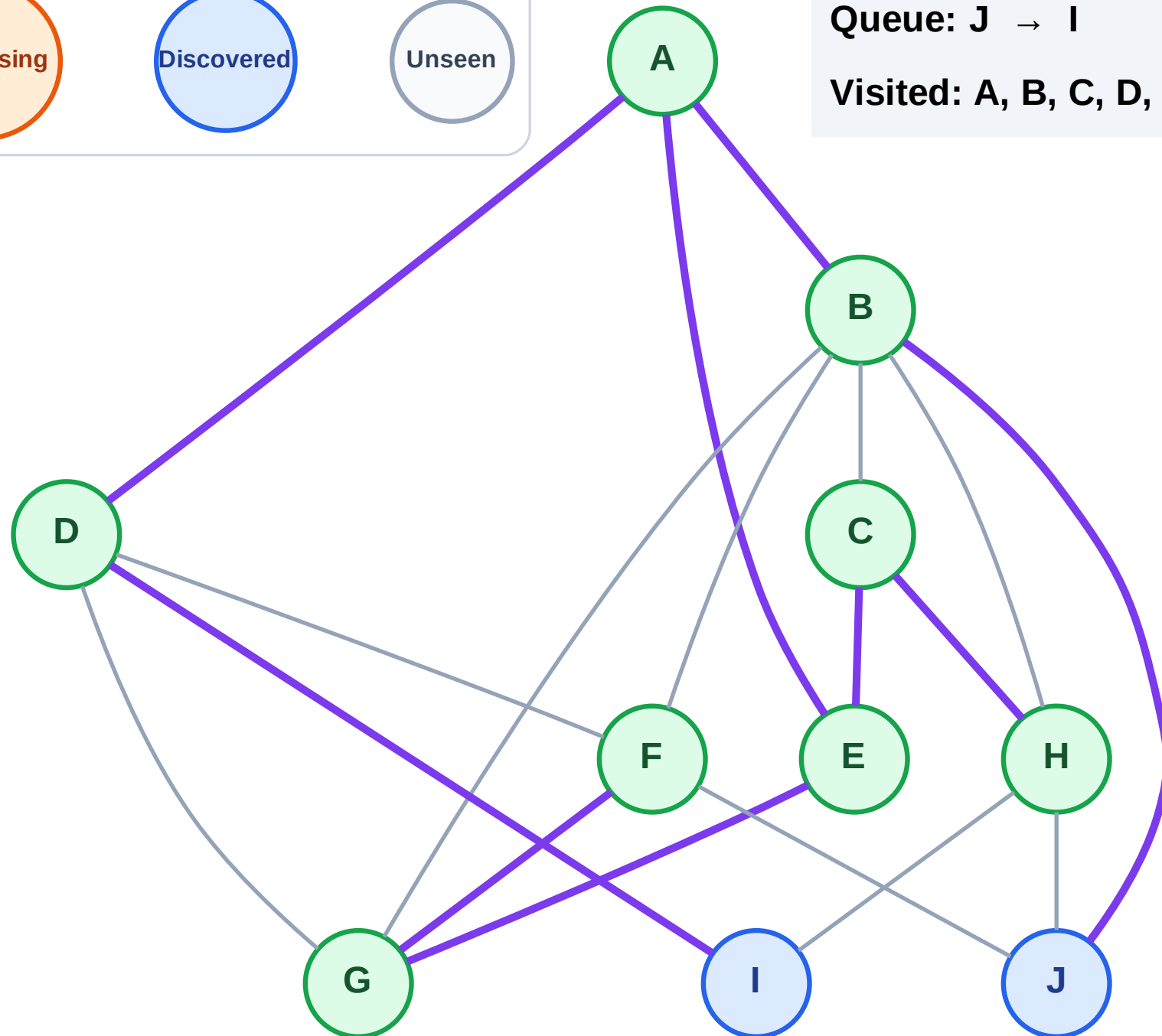
Step 17: No new neighbors from F

Legend



Queue: J → I

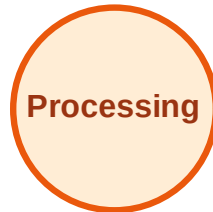
Visited: A, B, C, D, E, F, G, H



BFS Traversal

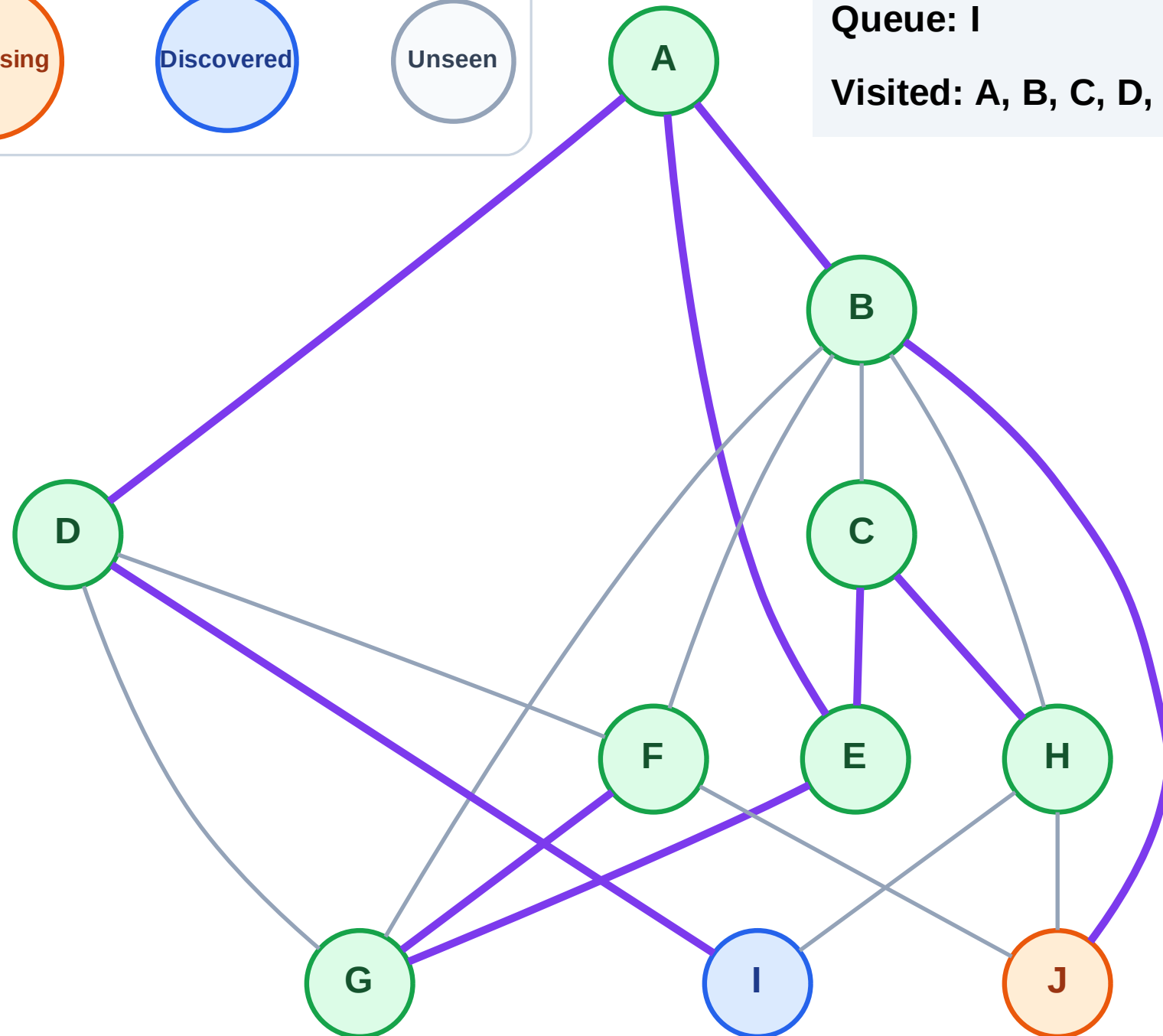
Step 18: Process node J

Legend



Queue: I

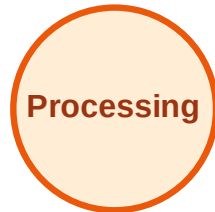
Visited: A, B, C, D, E, F, G, H



BFS Traversal

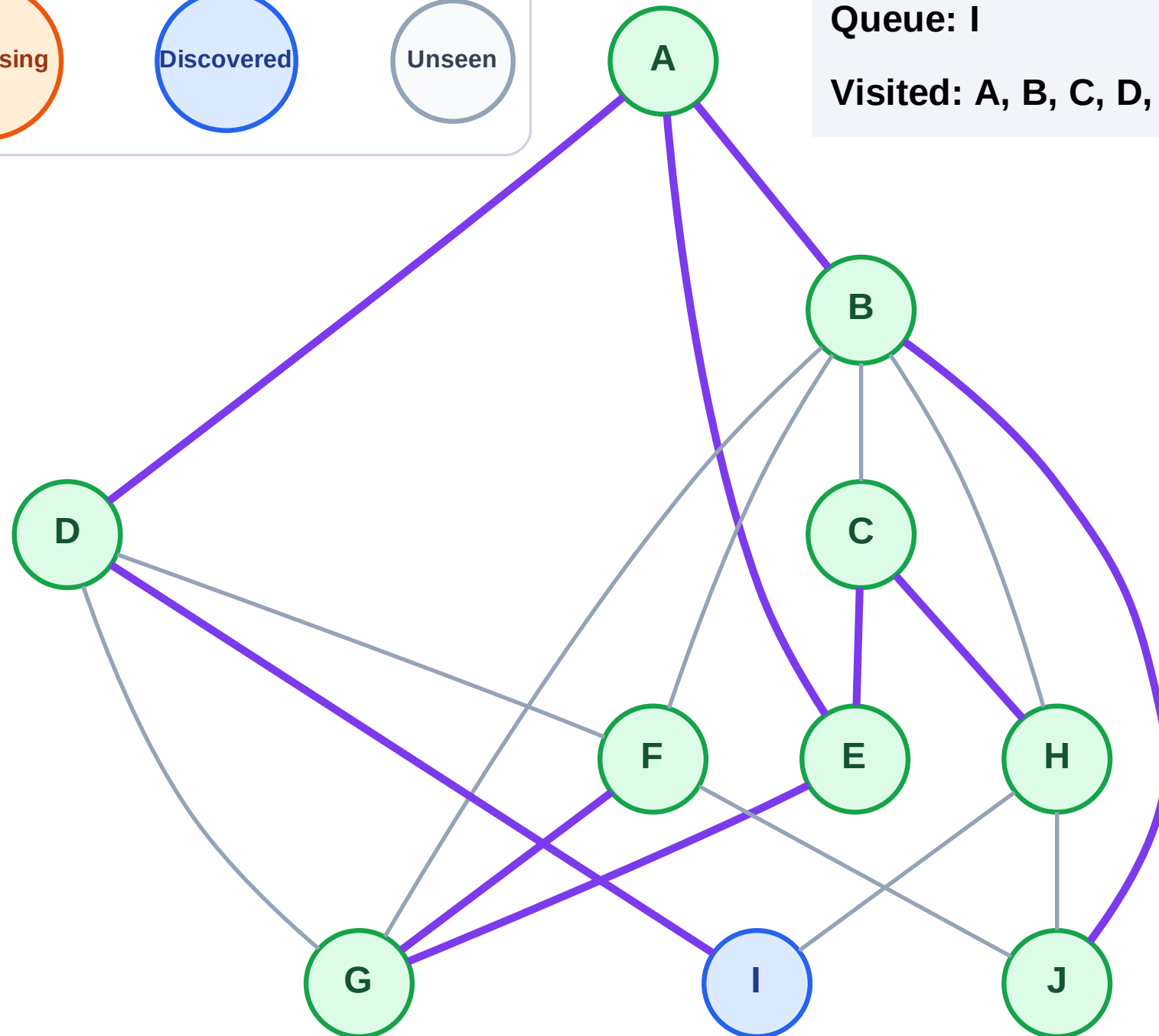
Step 19: No new neighbors from J

Legend



Queue: I

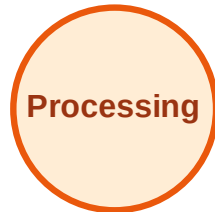
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

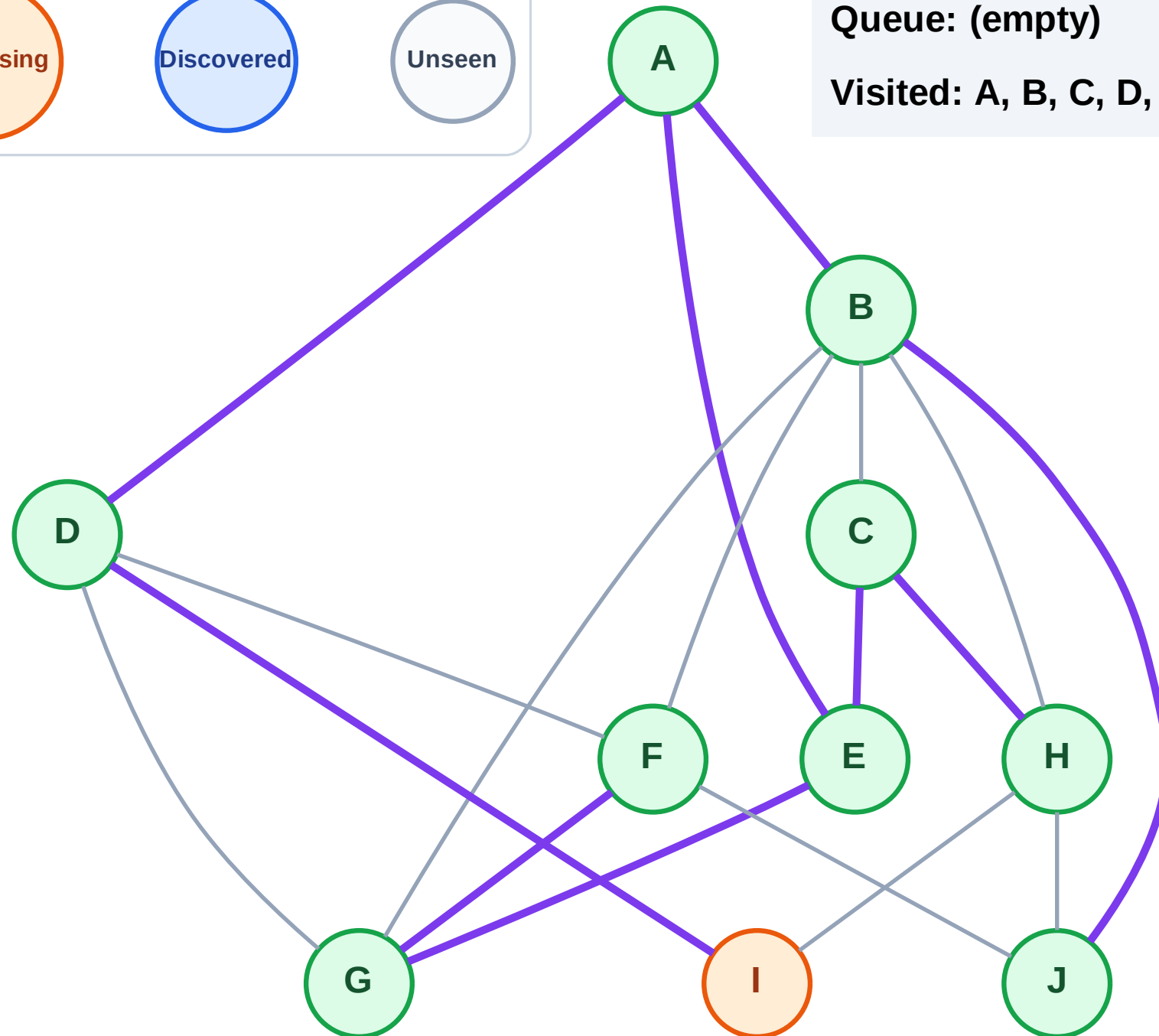
Step 20: Process node I

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, J

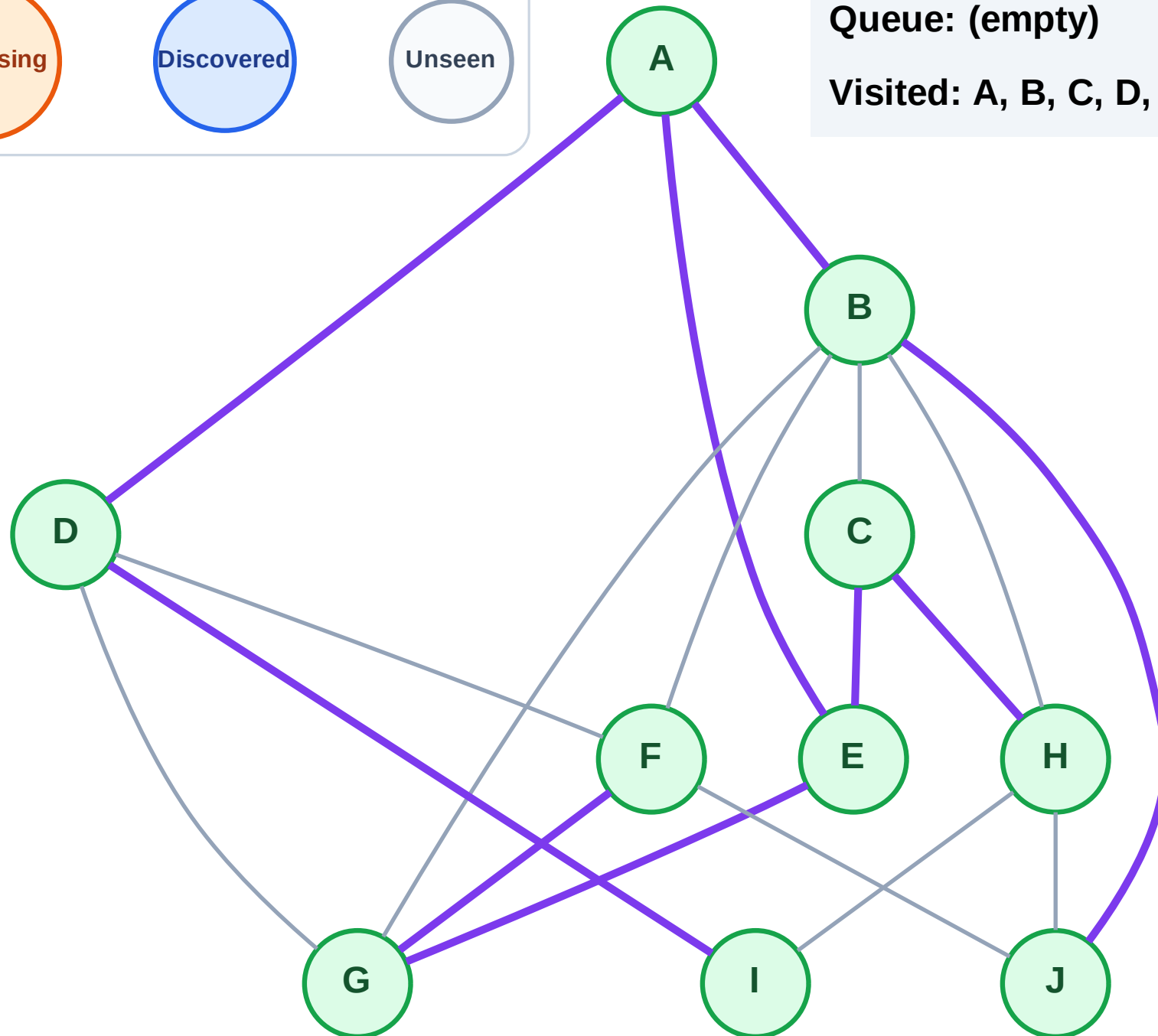


Step 21: No new neighbors from I

Legend



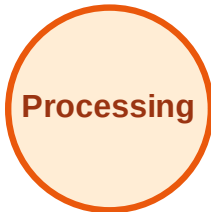
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

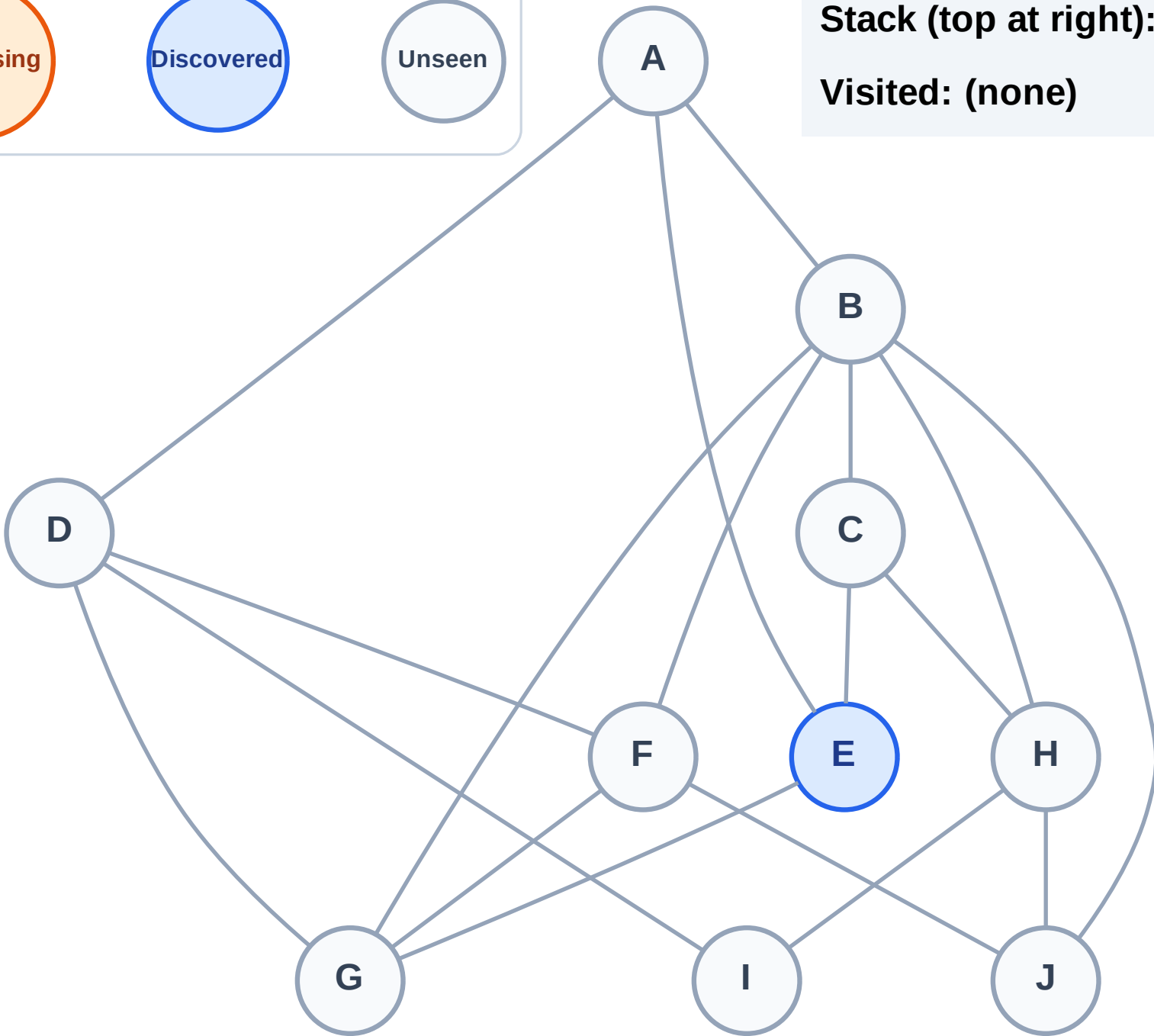
Step 1: Start at E

Legend



Stack (top at right): E

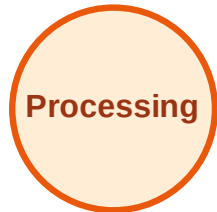
Visited: (none)



DFS Traversal

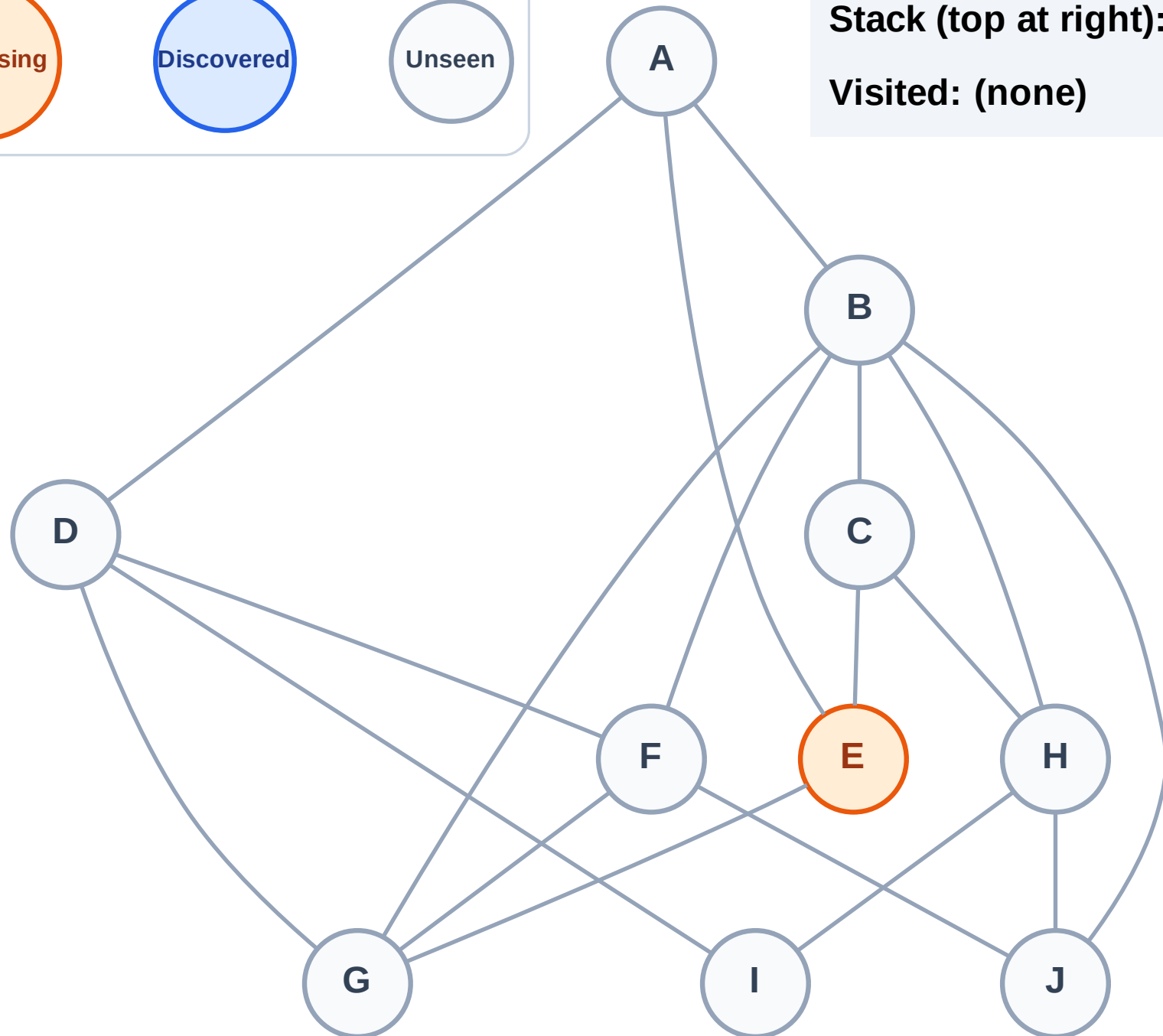
Step 2: Process node E

Legend



Stack (top at right): (empty)

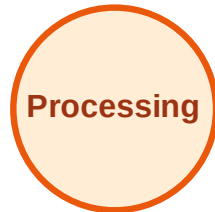
Visited: (none)



DFS Traversal

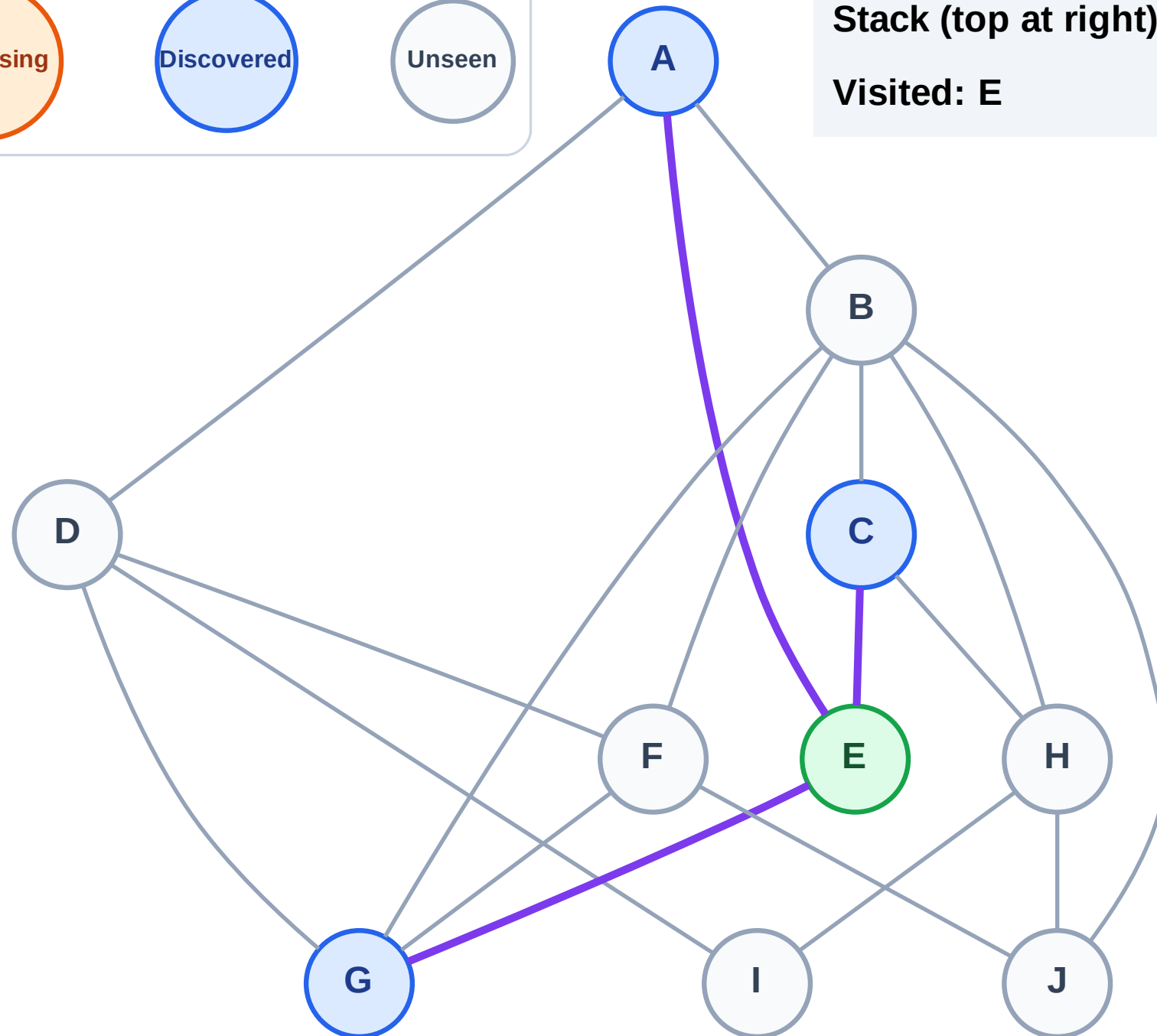
Step 3: Discover G, C, A from E

Legend



Stack (top at right): G | C | A

Visited: E



DFS Traversal

Step 4: Process node A

Legend

Complete

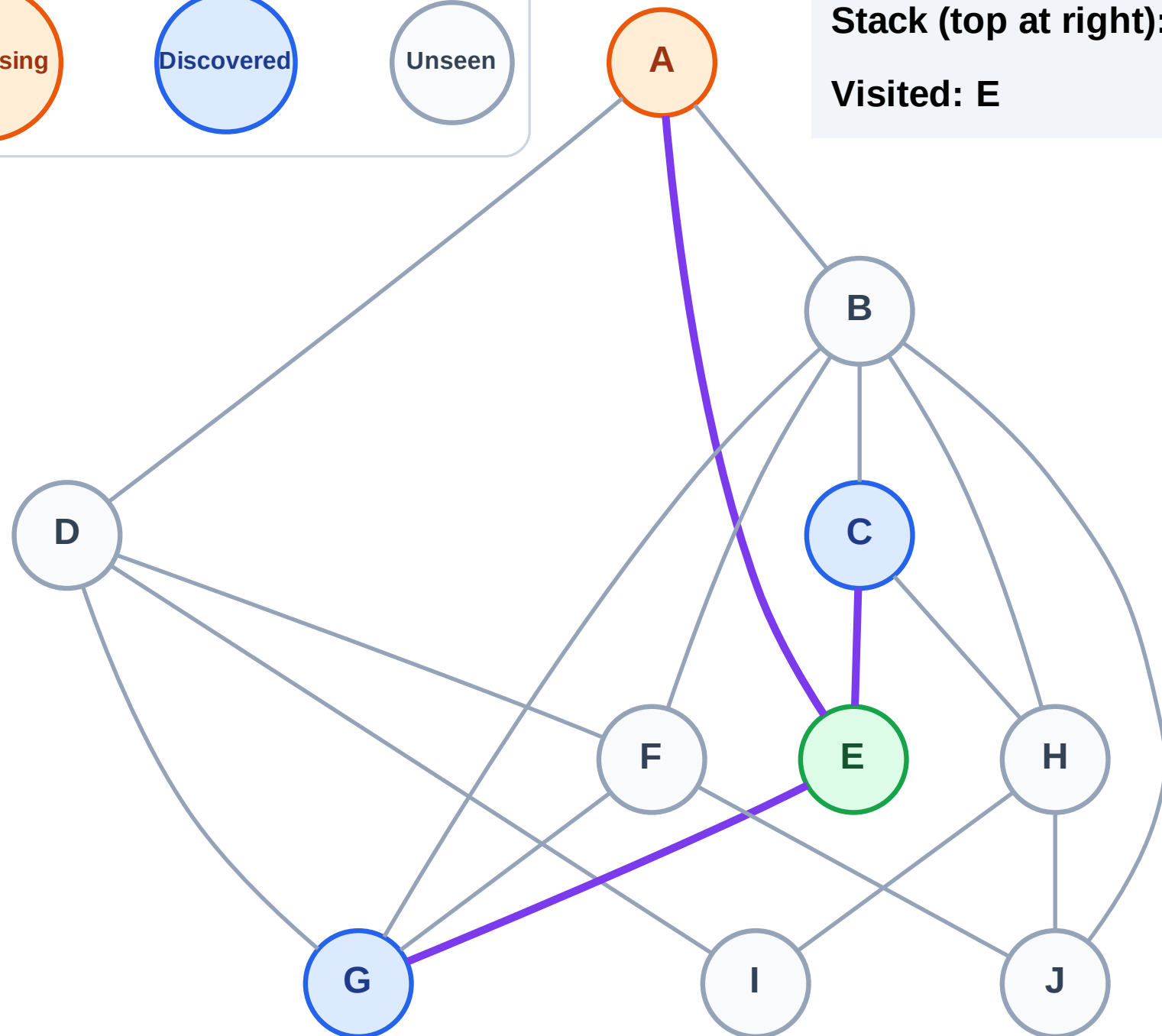
Processing

Discovered

Unseen

Stack (top at right): G | C

Visited: E



DFS Traversal

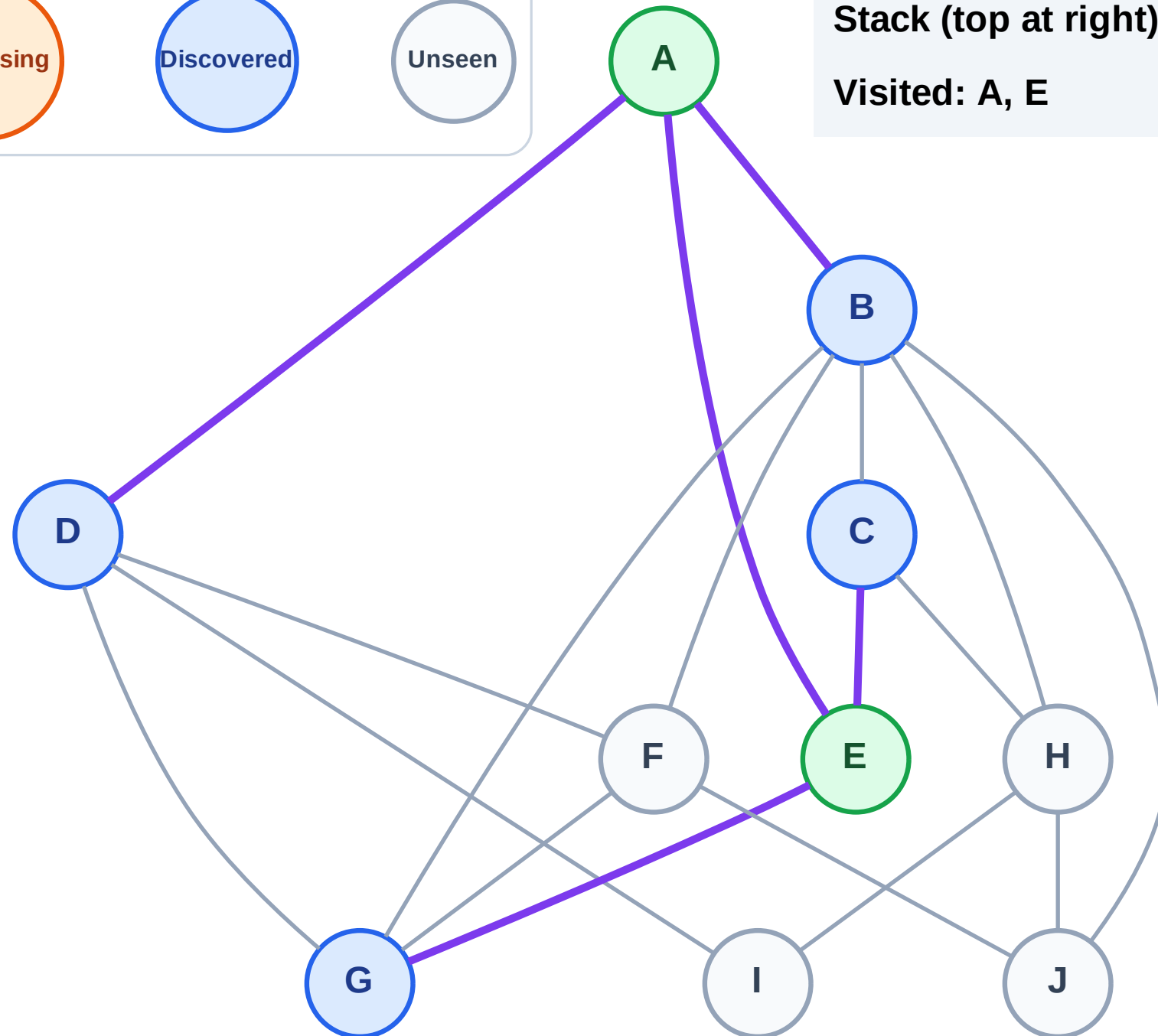
Step 5: Discover D, B from A

Legend



Stack (top at right): G | C | D | B

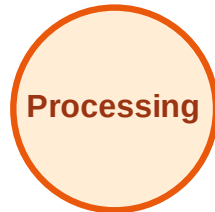
Visited: A, E



DFS Traversal

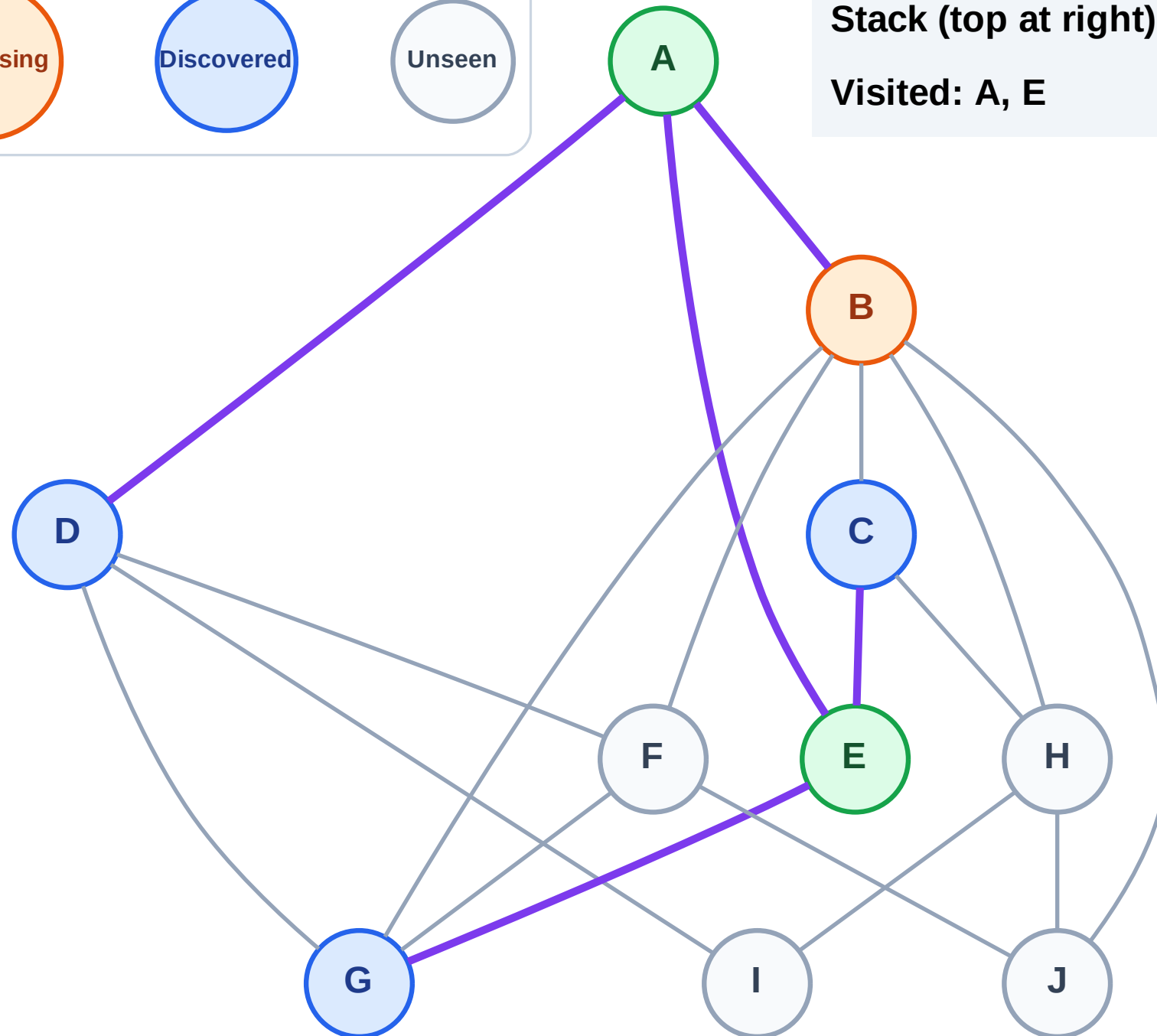
Step 6: Process node B

Legend



Stack (top at right): G | C | D

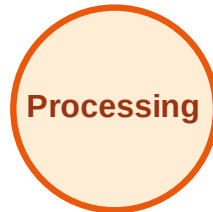
Visited: A, E



DFS Traversal

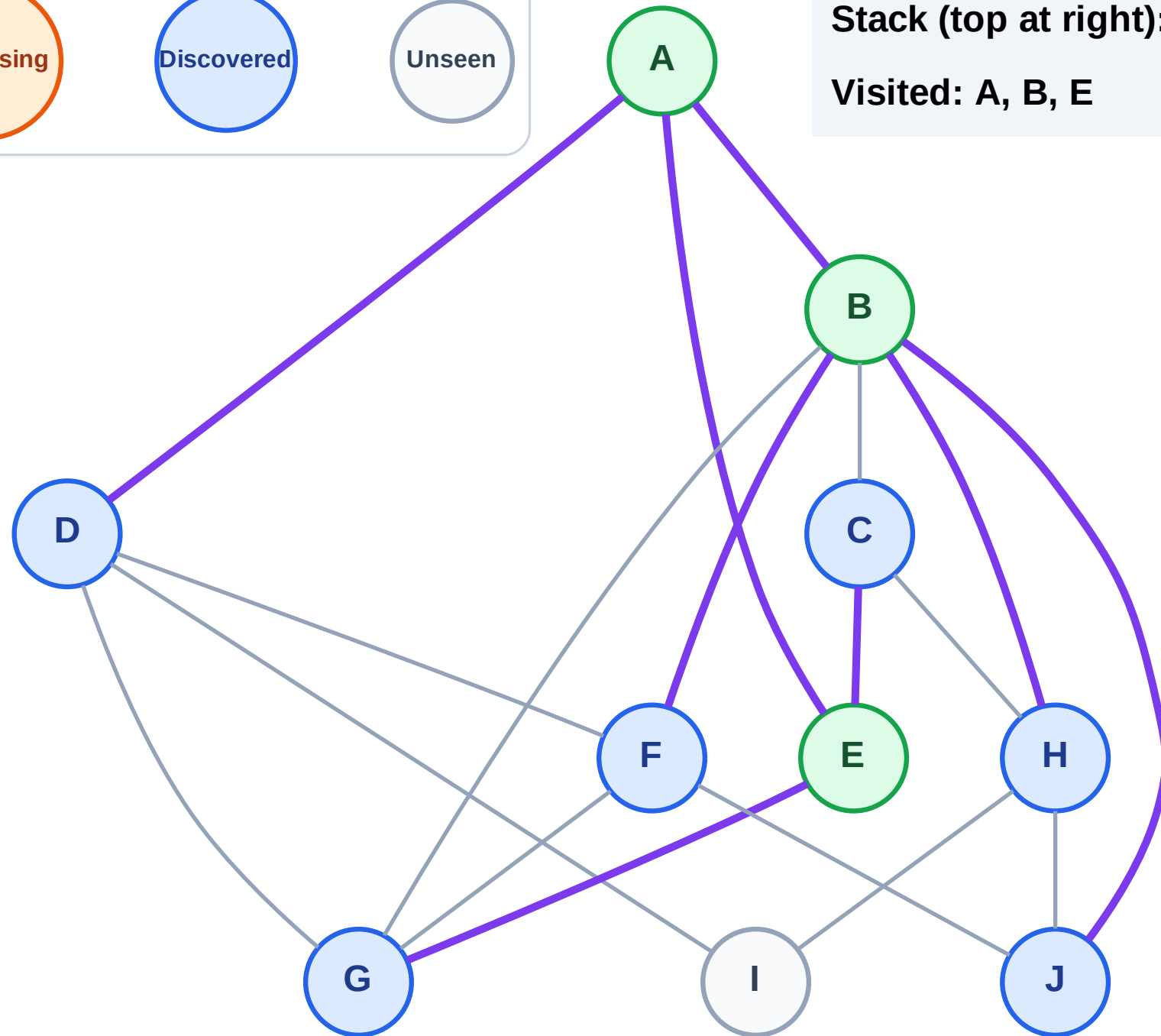
Step 7: Discover J, H, F from B

Legend



Stack (top at right): G | C | D | J | H | F

Visited: A, B, E



DFS Traversal

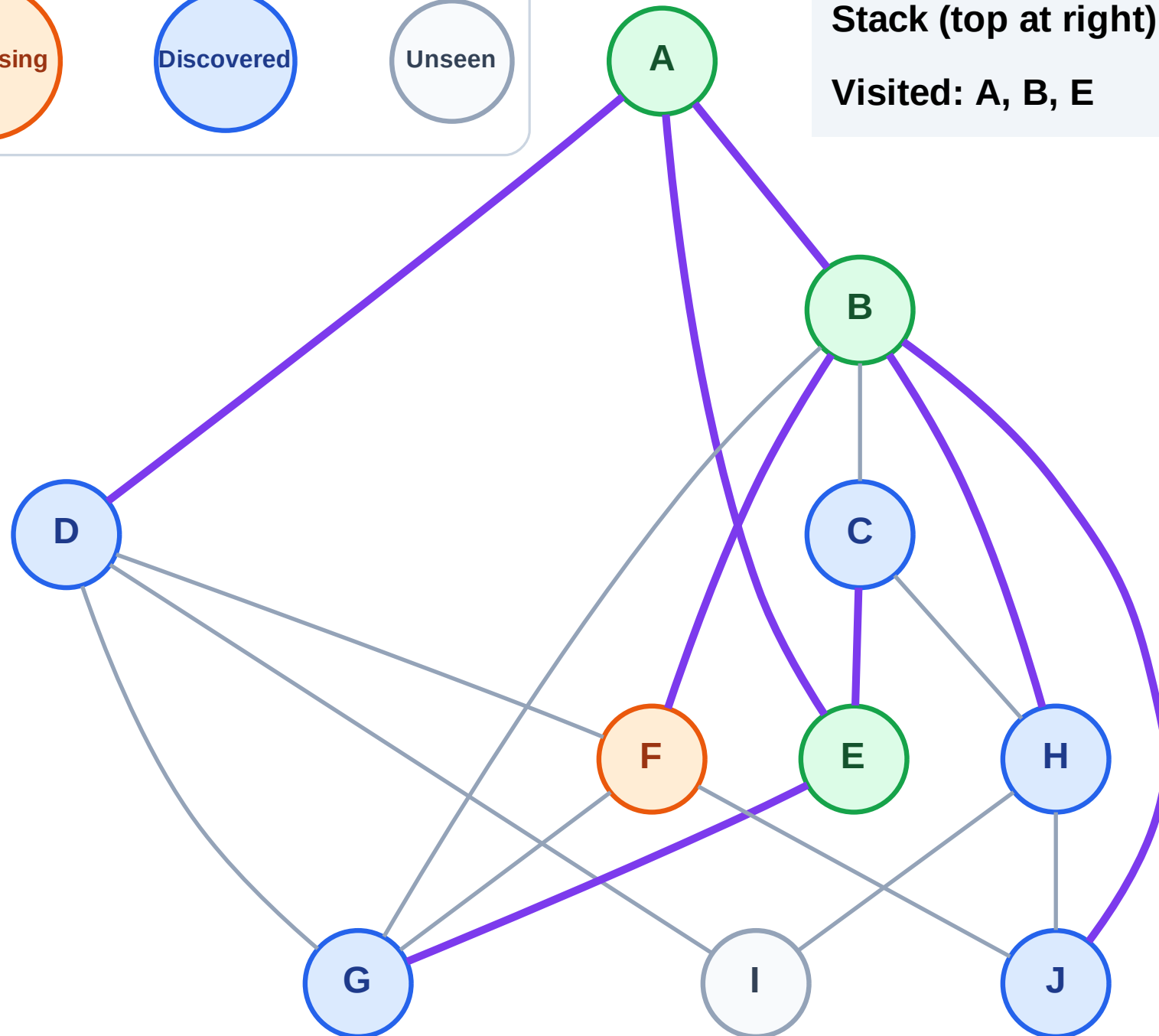
Step 8: Process node F

Legend



Stack (top at right): G | C | D | J | H

Visited: A, B, E



DFS Traversal

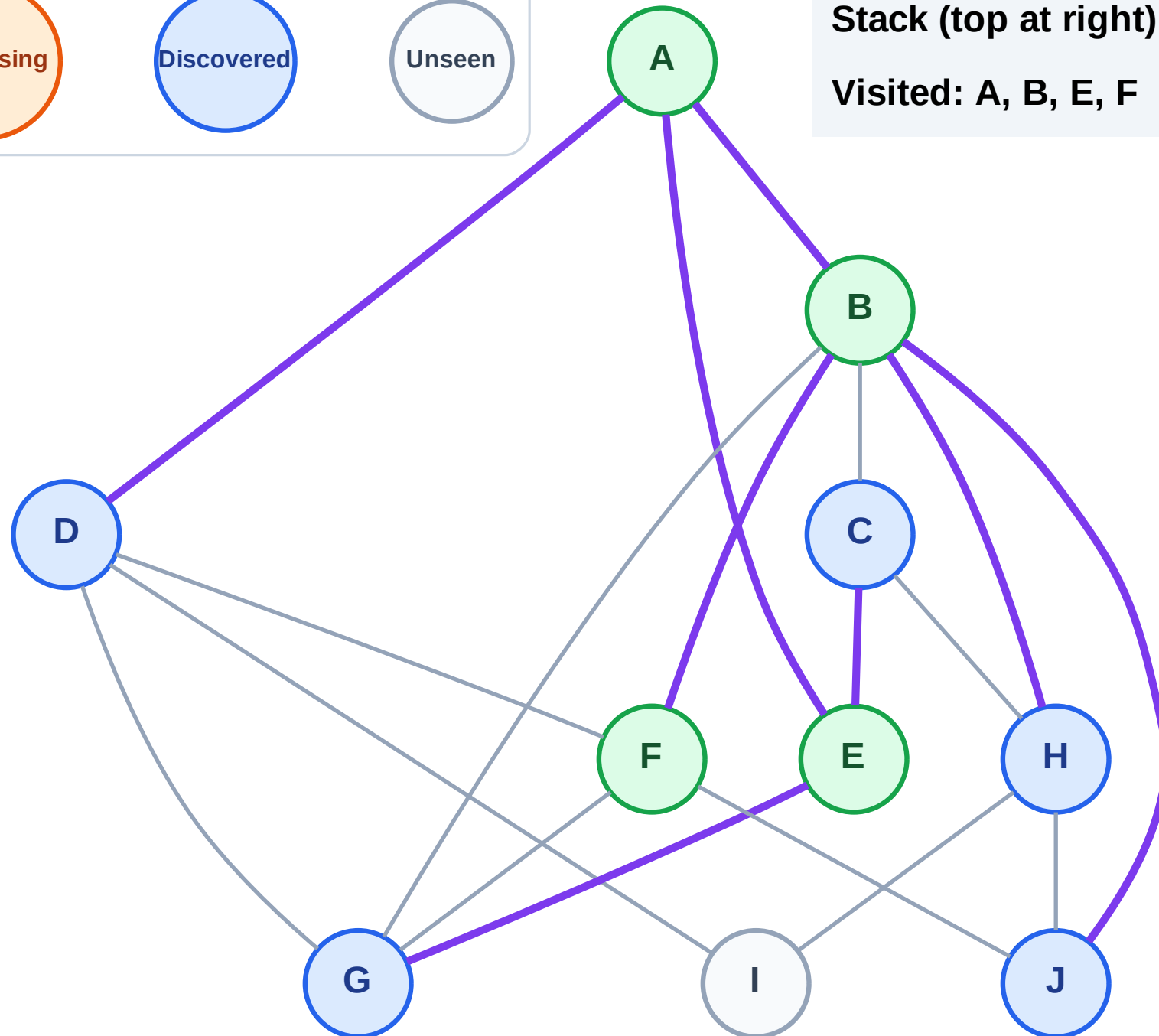
Step 9: No new neighbors from F

Legend



Stack (top at right): G | C | D | J | H

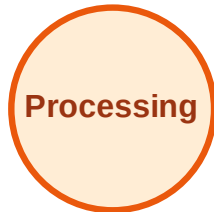
Visited: A, B, E, F



DFS Traversal

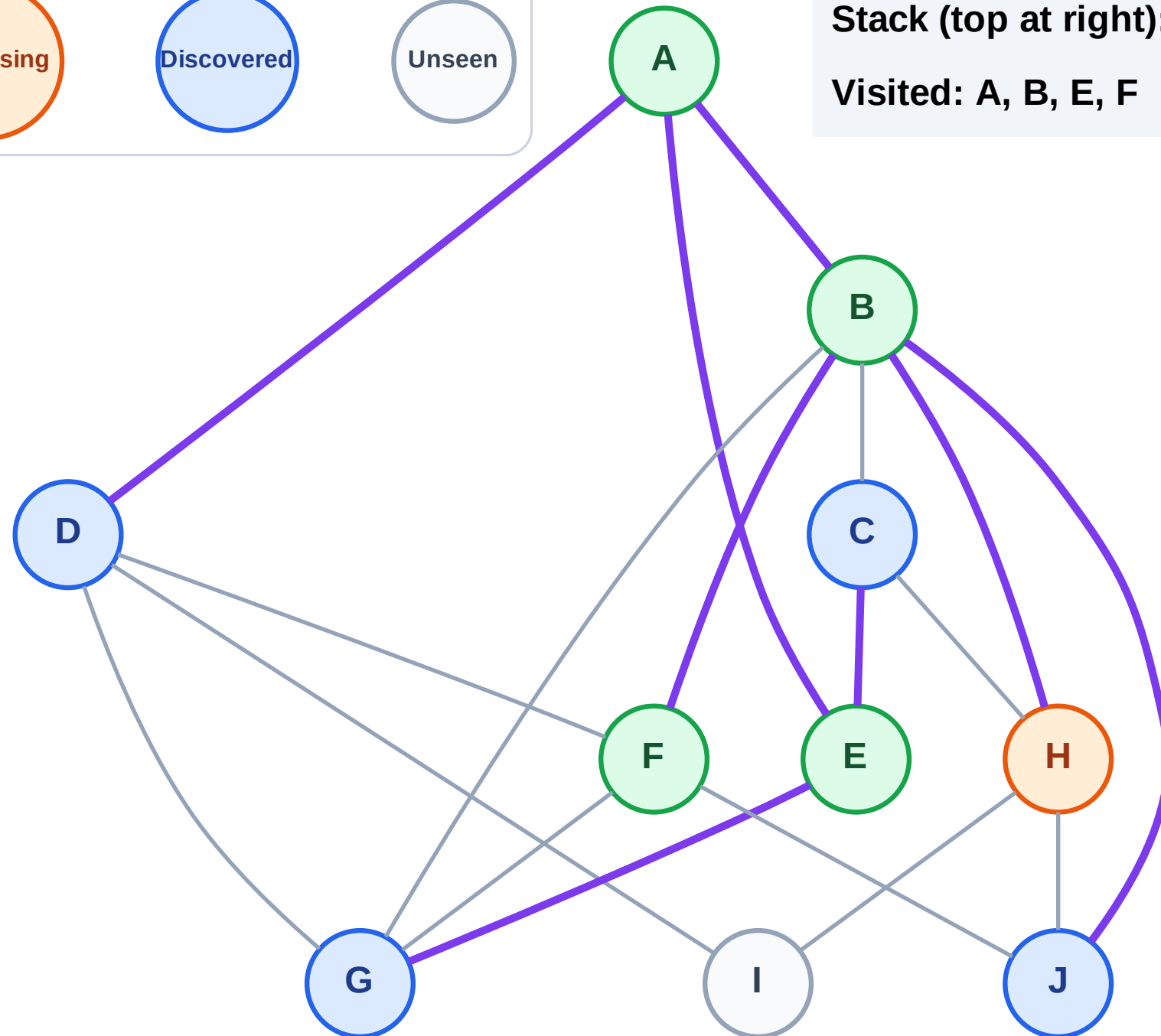
Step 10: Process node H

Legend



Stack (top at right): G | C | D | J

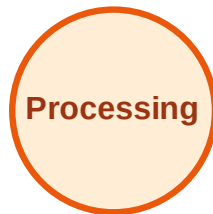
Visited: A, B, E, F



DFS Traversal

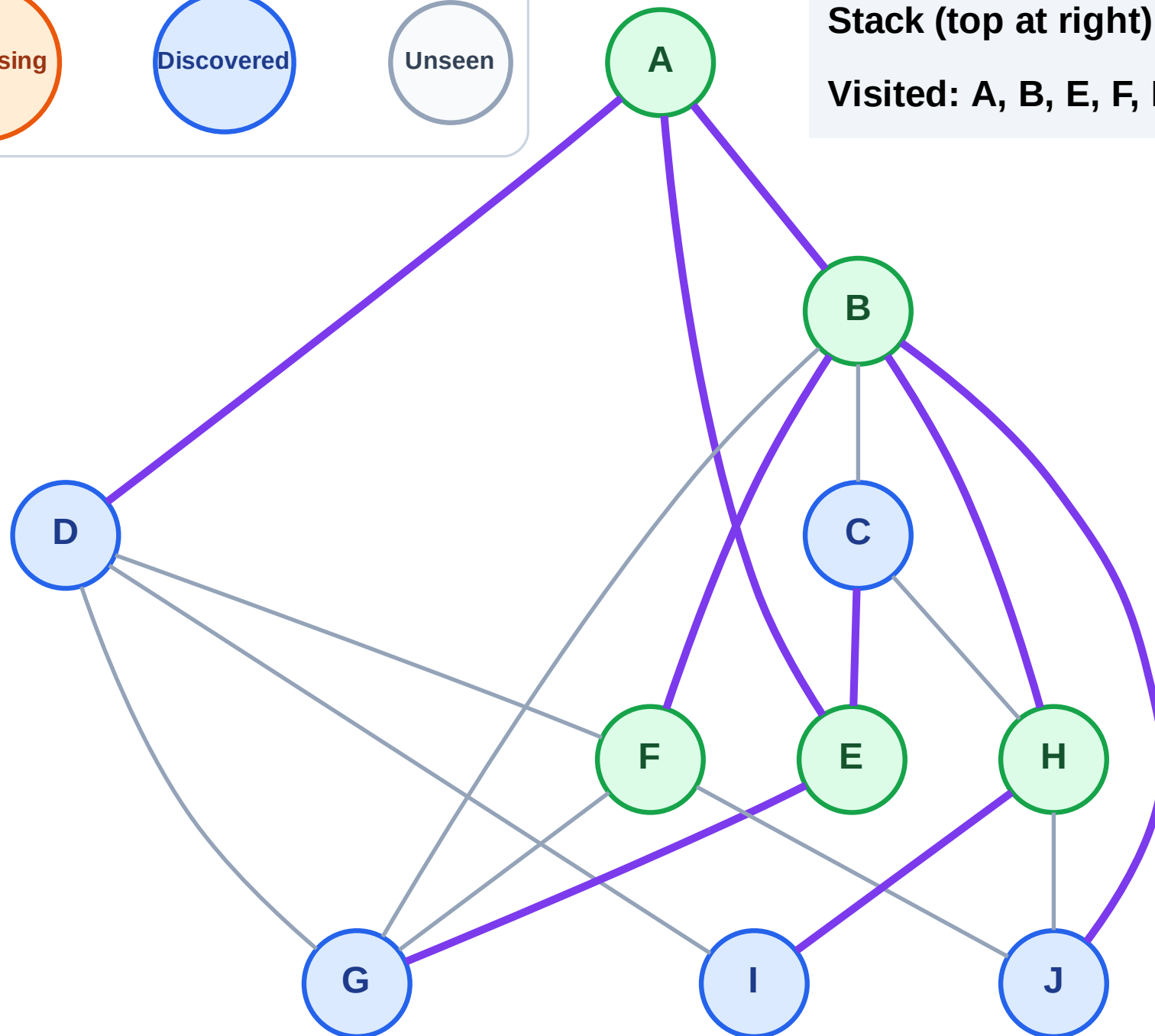
Step 11: Discover I from H

Legend



Stack (top at right): G | C | D | J | I

Visited: A, B, E, F, H



DFS Traversal

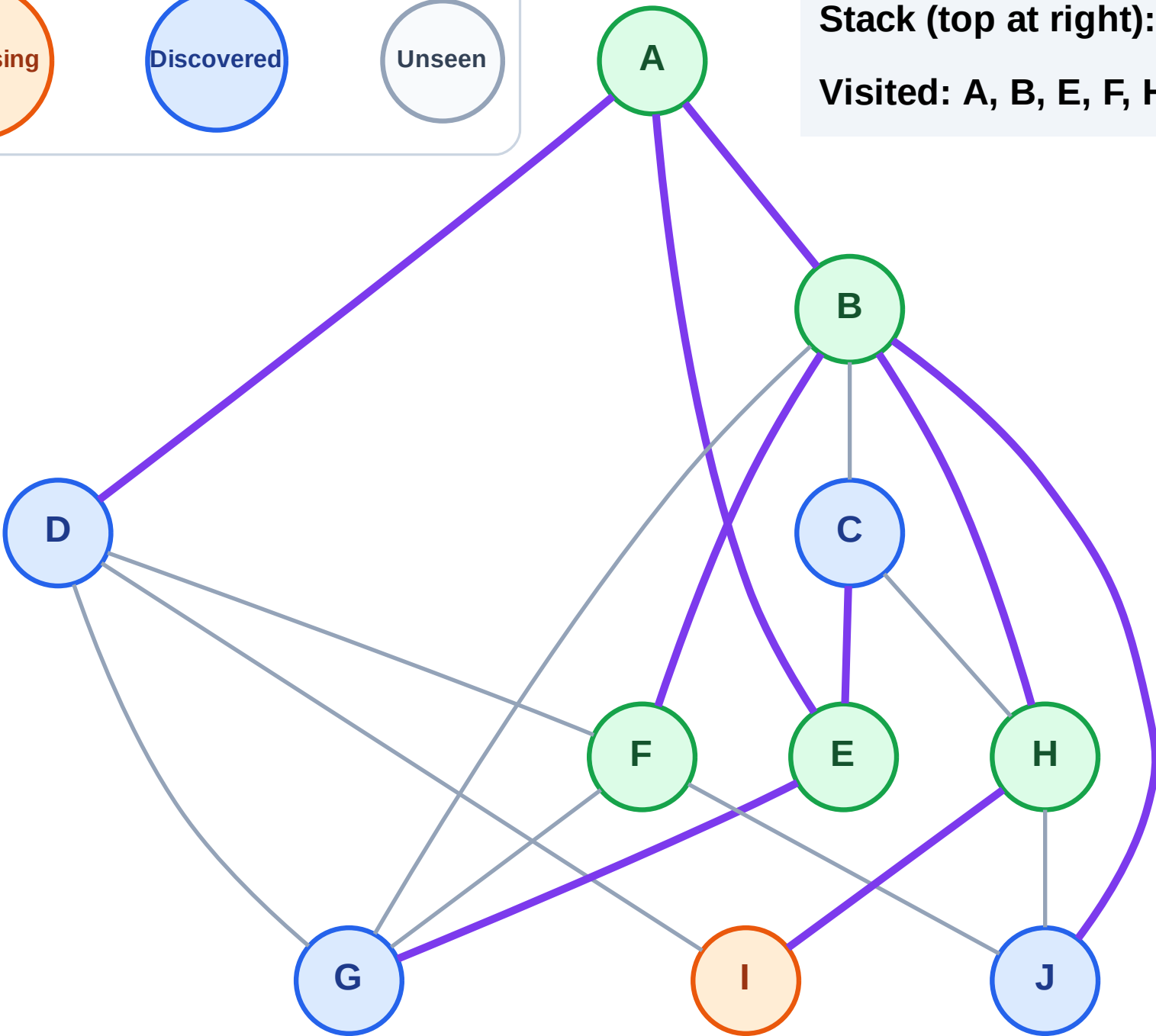
Step 12: Process node I

Legend



Stack (top at right): G | C | D | J

Visited: A, B, E, F, H



DFS Traversal

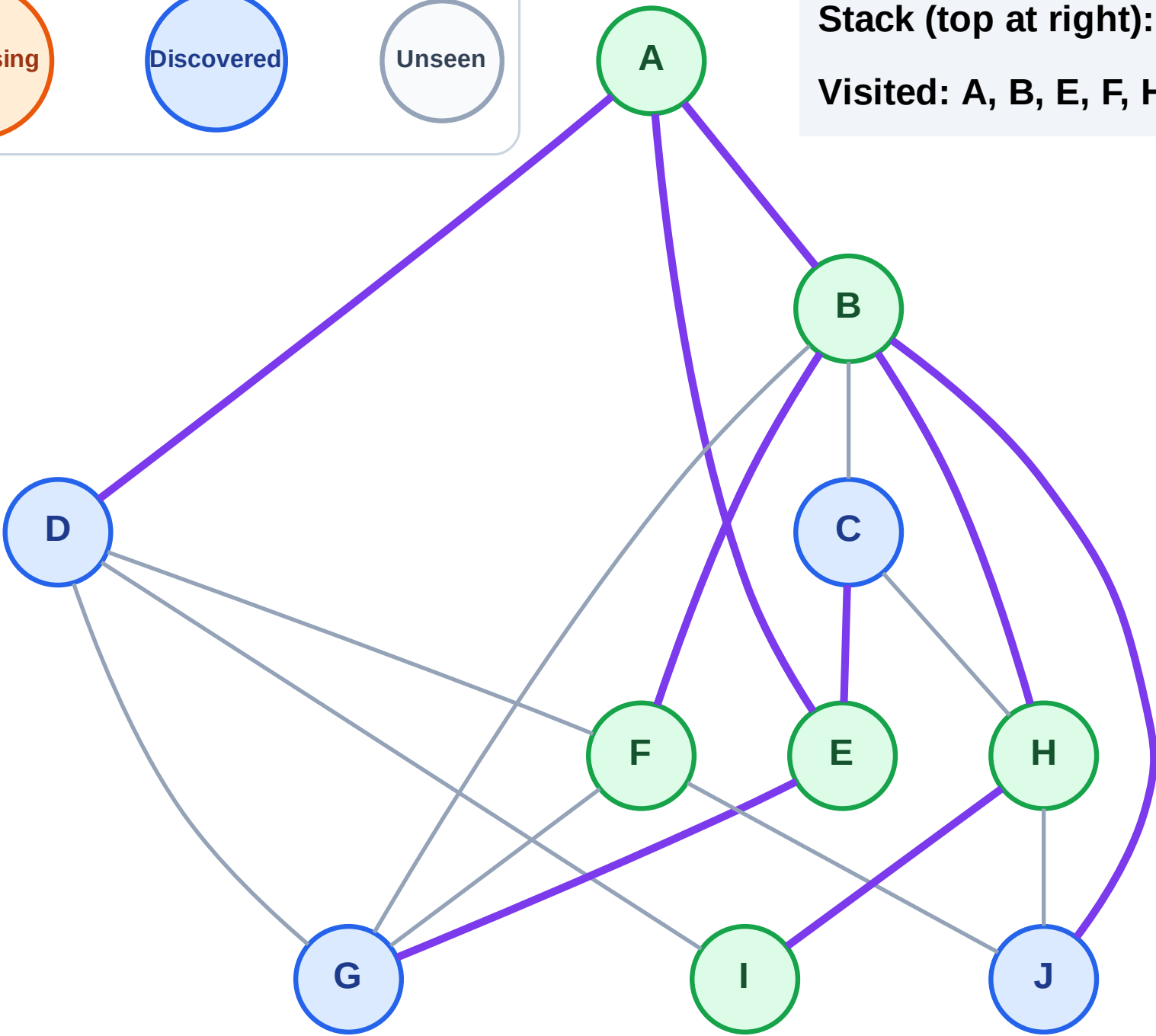
Step 13: No new neighbors from I

Legend



Stack (top at right): G | C | D | J

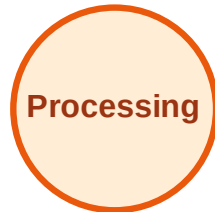
Visited: A, B, E, F, H, I



DFS Traversal

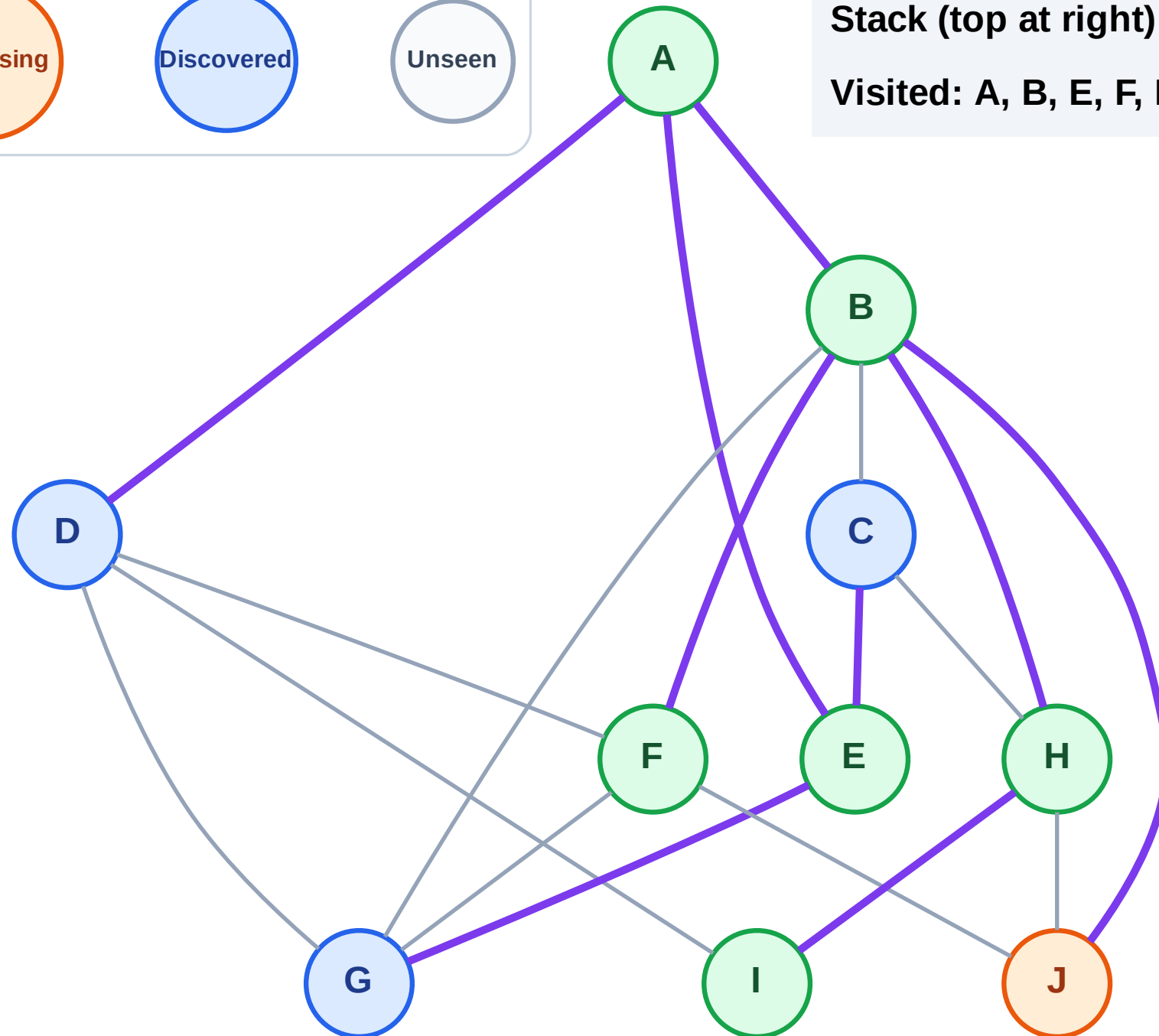
Step 14: Process node J

Legend



Stack (top at right): G | C | D

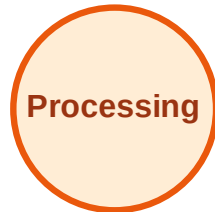
Visited: A, B, E, F, H, I



DFS Traversal

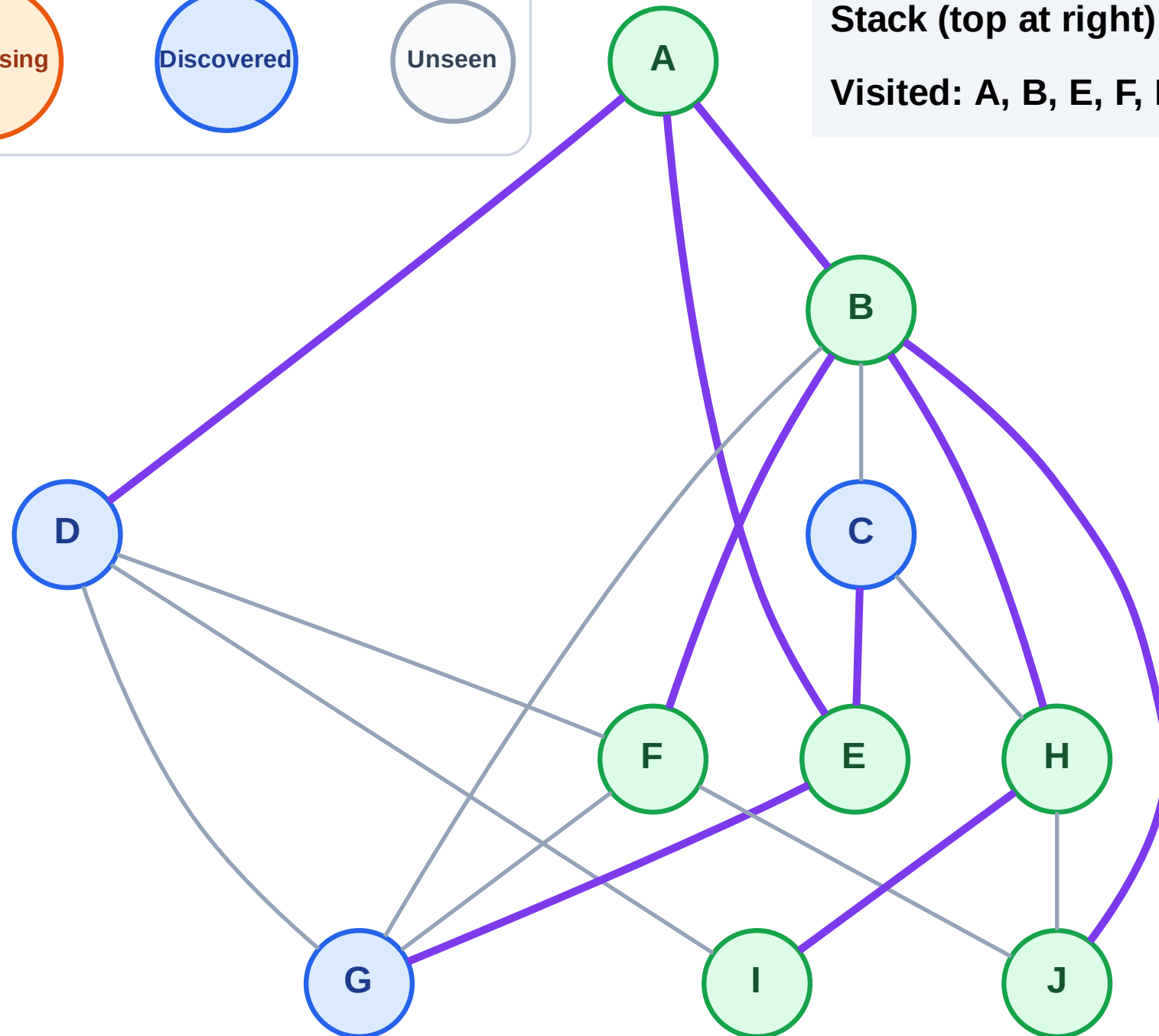
Step 15: No new neighbors from J

Legend



Stack (top at right): G | C | D

Visited: A, B, E, F, H, I, J



DFS Traversal

Step 16: Process node D

Legend

Complete

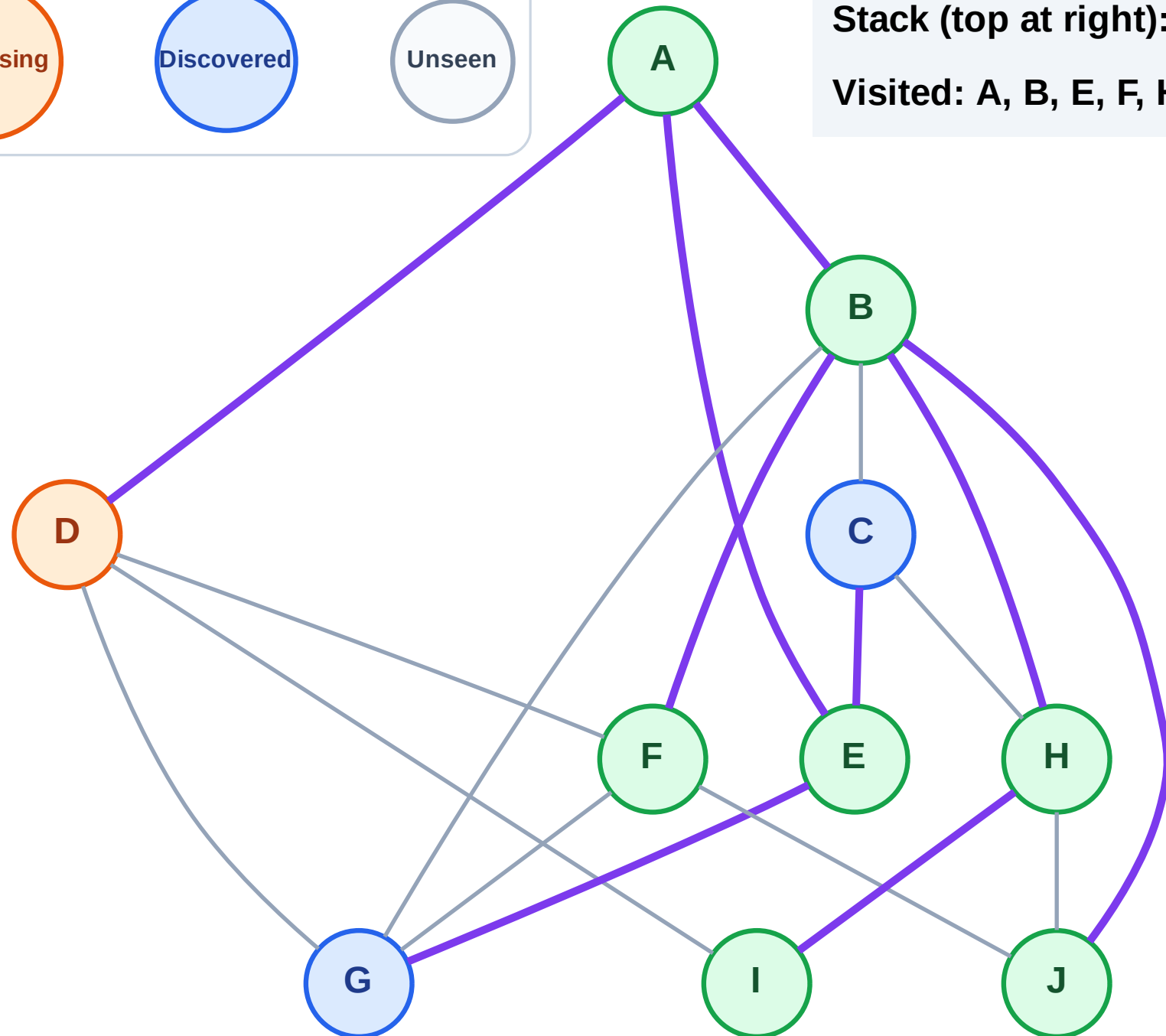
Processing

Discovered

Unseen

Stack (top at right): G | C

Visited: A, B, E, F, H, I, J



DFS Traversal

Step 17: No new neighbors from D

Legend

Complete

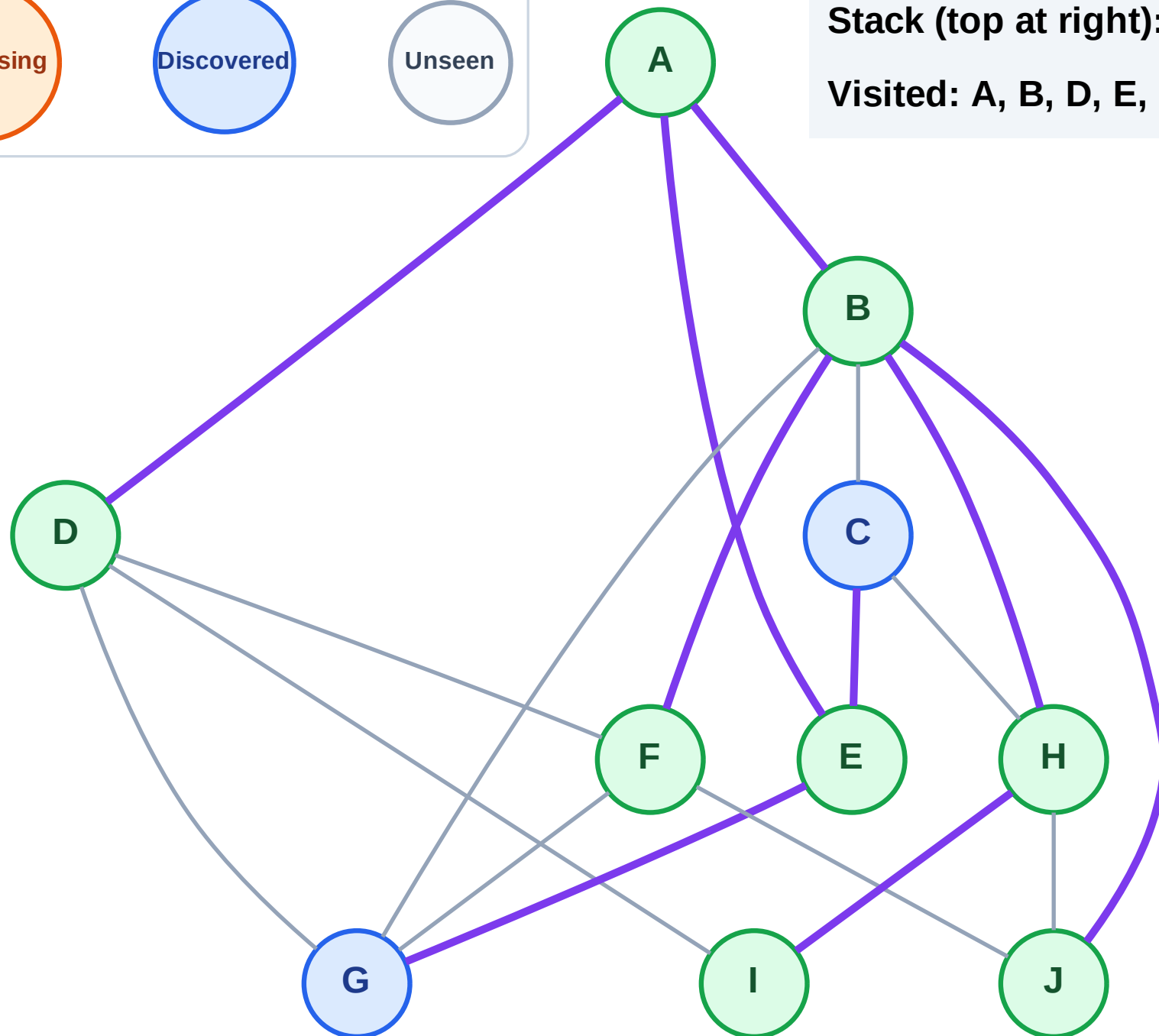
Processing

Discovered

Unseen

Stack (top at right): G | C

Visited: A, B, D, E, F, H, I, J



DFS Traversal

Step 18: Process node C

Legend

Complete

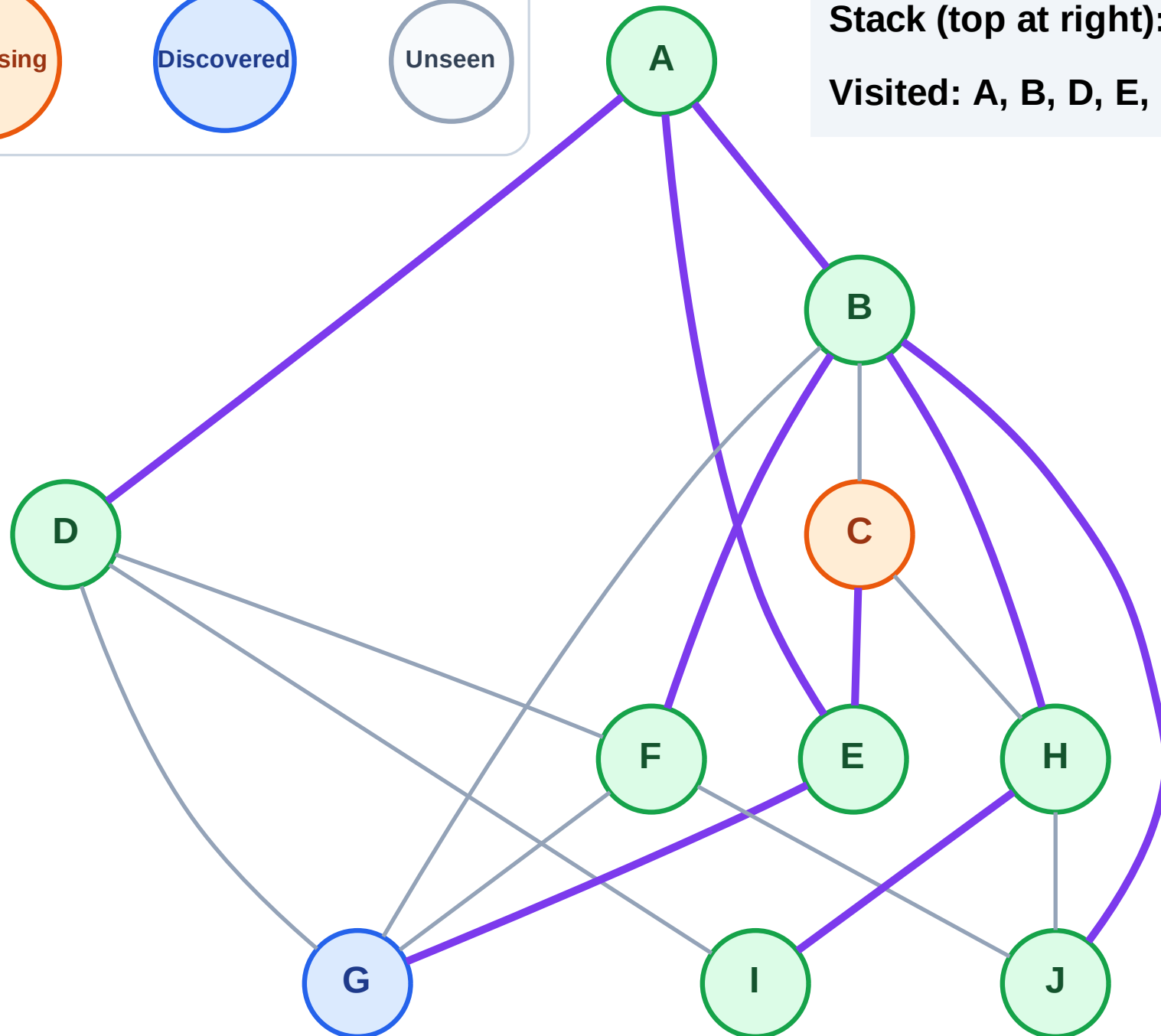
Processing

Discovered

Unseen

Stack (top at right): G

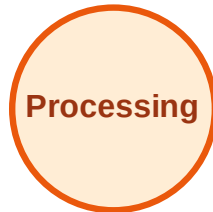
Visited: A, B, D, E, F, H, I, J



DFS Traversal

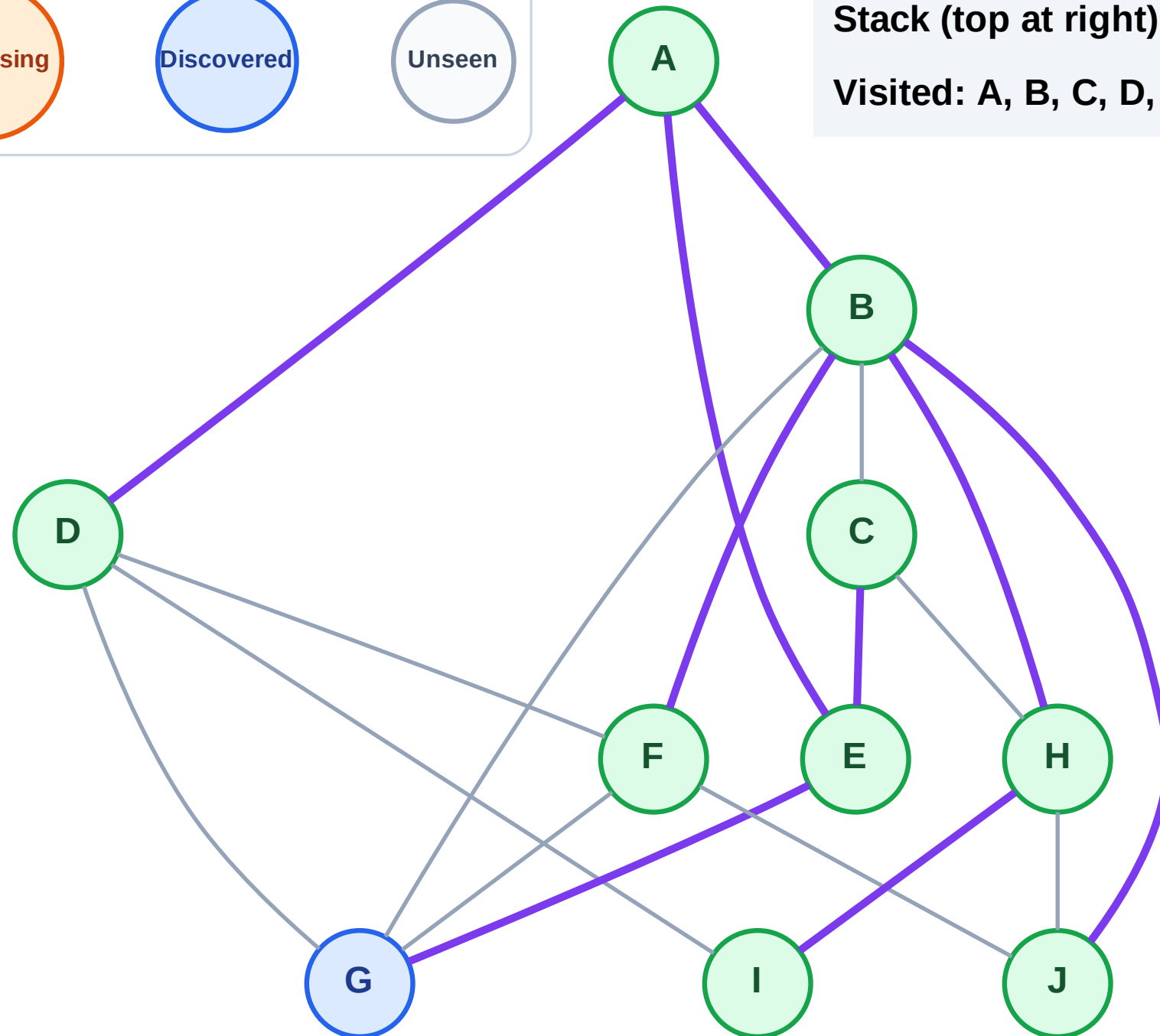
Step 19: No new neighbors from C

Legend



Stack (top at right): G

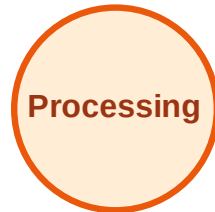
Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

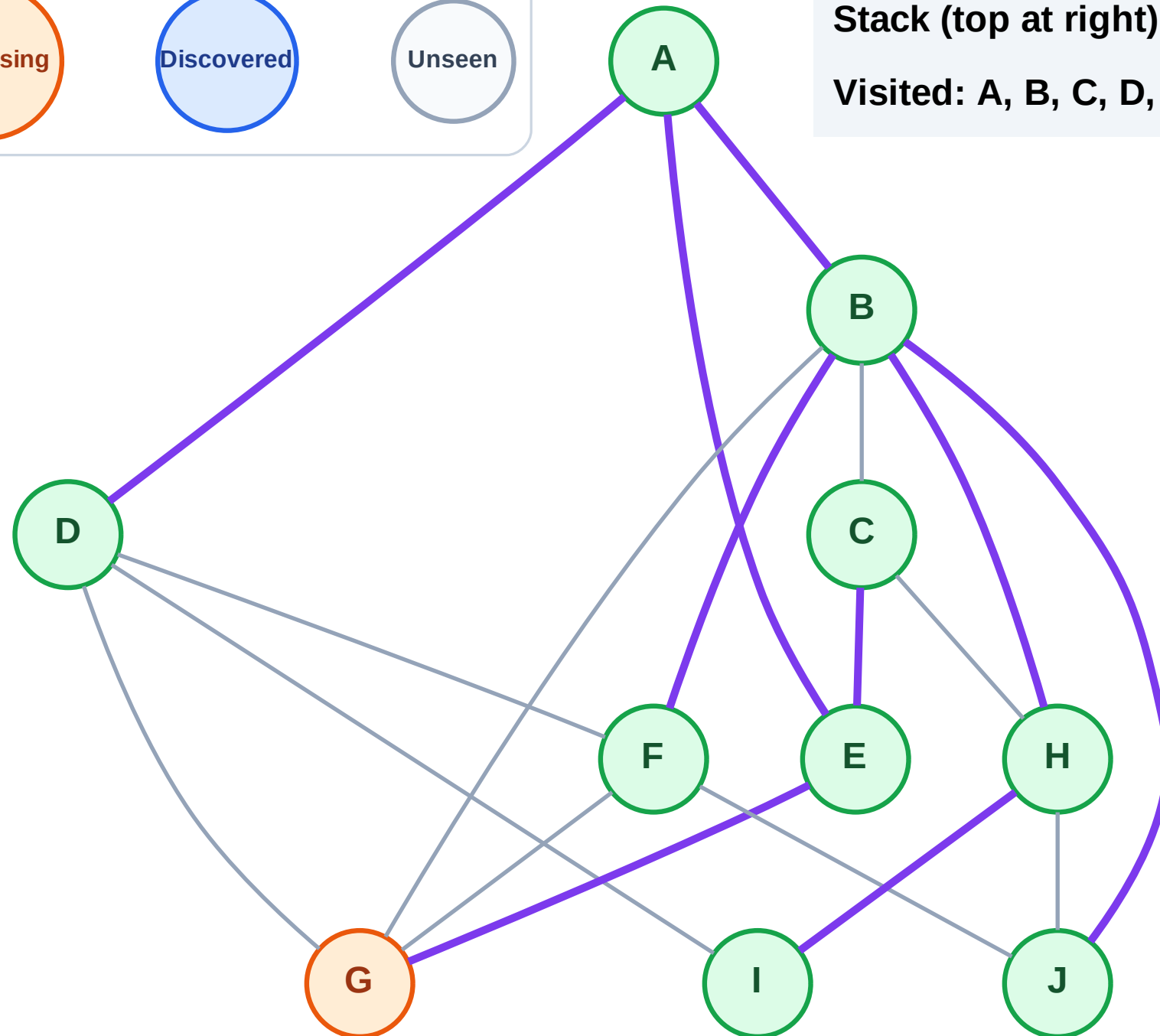
Step 20: Process node G

Legend



Stack (top at right): (empty)

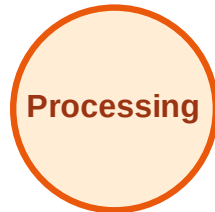
Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

Step 21: No new neighbors from G

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

